NTE: Network Topology Engine

A Hybrid Graph-Relational Model for Hyper-Scale Synthesis

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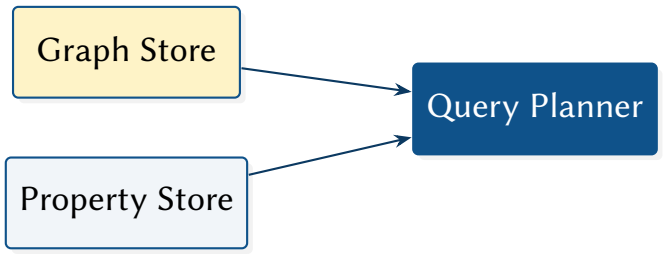

Abstract. NTE (Network Topology Engine) is a hybrid graph-relational engine for hyper-scale structural and property analytics in network topology modelling. Its central design principle is the Dual-Write Consistency paradigm, ensuring that topology structure and columnar properties are mutually isomorphic. Built to execute the Whiteboard-to-Plan (W2P) Strategy, this report documents the engine's "Correctness-by-Construction" policy engine, filter-first query planner, and the architectural foundations that underpin the AutoNetKit synthesis pipeline.

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Part I

Architecture & Insights

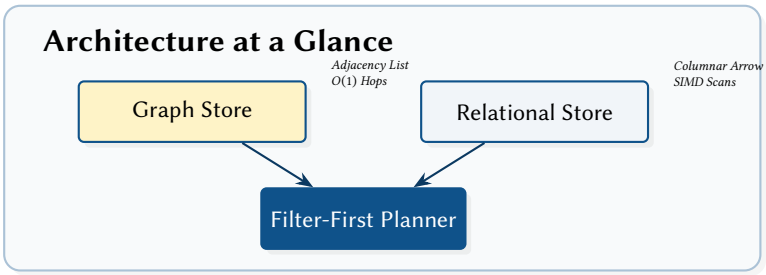
1 Introduction

The scale of modern infrastructure requires a fundamental shift in how topology data is stored and queried.

Insight: *Traditional graph databases excel at structural traversal but fail at columnar analytics; NTE bridges this gap by unifying both representations.*

: Dual-Write Consistency

Every structural mutation to the adjacency graph must be atomically reflected in the columnar property store. This isomorphism ensures that complex queries can be planned using either representation without risk of data divergence.



1.1 What NTE Solves

Network automation tools routinely need to answer two very different classes of question about a topology:

1. **Relational / property questions** — "Which routers have AS number 64512 and are in the SYD PoP?" — that are best answered by scanning columns of structured data.
2. **Graph / structural questions** — "Is there a path between device A and device B using fewer than 4 hops?" — that are best answered by traversing adjacency lists.

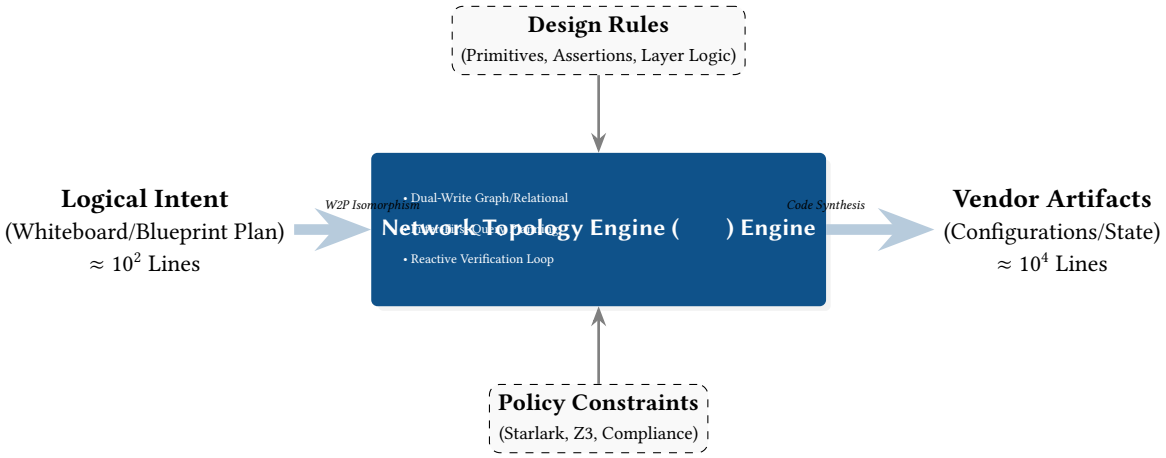


Figure 1. The Synthesis Funnel: Mapping Abstract Blueprint Intent to High-Fidelity Vendor Artifacts through the NTE Engine and formal Design Rules.

Most systems optimise for one and bolt on the other as an afterthought. Relational databases support graph queries via recursive CTEs that are notoriously hard to optimise. Dedicated graph databases (Neo4j [1], KuzuDB [2]) support property lookups but typically store properties row-by-row and lack columnar scan performance.

NTE's answer is a *dual-write* architecture: every mutation is applied to both a petgraph `StableDiGraph` (for structural queries) and a set of Polars Arrow-backed DataFrames (for property scans). A filter-first query planner decides, at runtime, which representation to consult first. In the common case — filter on a property, then traverse — it runs the cheap columnar scan to get a candidate set of node IDs, then constrains the traversal to those candidates.

1.2 Why a Bespoke Engine?

A natural question is: *Why not use an off-the-shelf graph database like Neo4j, ArangoDB, or Memgraph?* While excellent systems, traditional graph databases impose significant overhead for network automation:

- **Network overhead:** Executing millions of small structural validations via a wire protocol (e.g., Bolt) introduces unacceptable latency. NTE embeds directly in the Python process via PyO3, communicating via zero-copy Arrow memory.
- **Property scan performance:** Most graph DBs store properties row-by-row. When querying "all interfaces with speed > 100G", they cannot match the SIMD-accelerated columnar scan speeds of Polars.
- **Deterministic compilation:** The `ank_netcfg` compiler requires a strictly deterministic, pure-Rust graph engine to ensure reproducible configuration generation without relying on an external daemon.

Table 1. Architectural Positioning: NTE vs. State-of-the-Art.

Feature	Neo4j	PostgreSQL	Batfish	NTE
Primary Model	Graph (Row)	Relational (Row)	Simulation	**Hybrid (SIMD/Graph)**
Storage Engine	Native Graph	Heap Tables	In-memory IR	**Arrow/CSR Dual-Write**
Query Primary	Cypher	SQL (CTE)	Declarative	**Filter-First SIMD**
FFI Overhead	High (RPC)	High (RPC)	N/A (Tool)	**Zero (Embedded PyO3)**
"What-If" Speed	Medium (Clone)	Slow (Snapshot)	Fast	**Instant (CoW Shadow)**

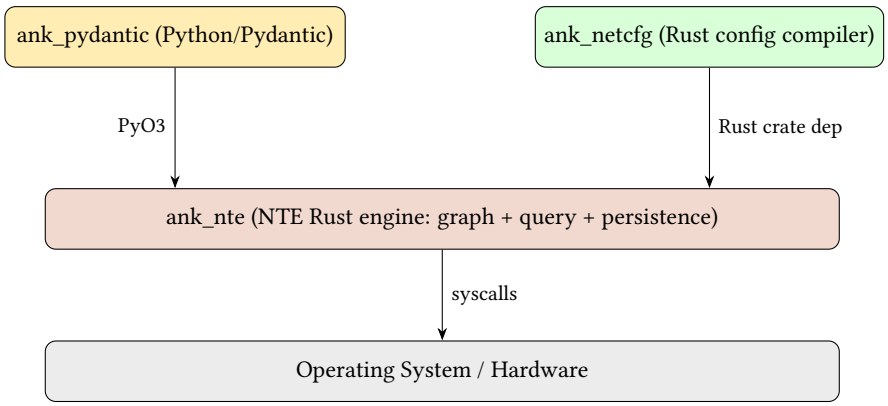
1.3 Ecosystem: ank_pydantic and ank_netcfg

NTE is a shared backend engine consumed by two alternative frontends:

`ank_pydantic` A Python library that provides high-level, Pydantic-validated network models. It owns the domain schema (what a *Router* looks like, what fields are mandatory) and calls NTE via PyO3 bindings. Suited to interactive exploration, Jupyter notebooks, and Python-centric automation pipelines.

`ank_netcfg` A pure-Rust deterministic network configuration compiler. It consumes `nte-topology` directly (no Python layer), reads declarative YAML blueprints, maps them through a DeviceIR intermediate representation, and renders vendor-neutral device configurations. Suited to CI/CD pipelines, large-scale config generation, and environments where a Python runtime is undesirable.

Both frontends use the same underlying NTE storage, query, and persistence layers. The choice between them is an ergonomic decision, not an architectural one.



All three repositories live side-by-side in the same parent directory.

1.4 Whiteboard-to-Plan (W2P) Strategy

The fundamental architectural principle of NTE is the **Whiteboard-to-Plan (W2P)** strategy. Rather than building a topology through low-level imperative mutations, high-level visual intent (from a whiteboard or diagram) is aggregated into a standardized **Blueprint Plan**.

: The W2P Isomorphism

Every visual primitive ω in a whiteboard domain must map to a formal assertion α in the Blueprint Plan. The mapping function $\phi : \omega \to \alpha$ is a deterministic transformation that preserves the semantic intent of the operator while expanding it into its constituent NTE nodes and properties.

This strategy decouples the *creative intent* (the network design) from the *structural implementation* (the graph/relational storage). A Blueprint Plan acts as a contract: once the visual intent is "frozen" into a plan, the NTE engine maps it to the layered topology by applying a set of **Design Rules** (Primitives and Assertions).

This ensures that the final synthesised topology is not merely a collection of nodes, but a provably correct representation of the original design intent, validated at every stage of the "Synthesis Funnel" (see Figure 1).

§§2–4 cover the core data model and storage engine. §5 covers the query system: the filter-first planner, the fluent Python query builder, NTE-QL, and the interactive REPL. §7 covers ACID transactions and conflict detection. §§8–13 cover advanced topics — persistence, GPU acceleration, graph algorithm plugins, policy engine (including shadow topologies), reactive events, production diagnostics, and Raft clustering — that can be read in any order.

2 Data Model

Source

Axiom 1 (Dual-Write Atomicity). *Every mutation to the topology must be reflected atomically in both the graph adjacency store and the columnar property store. A transaction is only committed if both representations are successfully updated.*

Theorem 1 (Graph-Relational Equivalence). *The set of all nodes V and edges E reachable via graph traversal is equivalent to the set of rows in the primary Polars DataFrames. $\forall v \in DiGraph, \exists! row \in DataFrame$ mapping to v.*

2.1 Formal Definition

A topology in NTE is a typed, directed, attributed multi-graph $G = (V, E, \ell_V, \ell_E, \pi_V, \pi_E, L, \lambda)$ where:

$$
\begin{aligned}
V &= V_{\text{node}} \cup V_{\text{endpoint}} && \text{(disjoint node and endpoint sets)} \\
E &\subseteq V \times V && \text{(directed edges)} \\
\ell_V &: V \to \Sigma_V && \text{(type label, e.g. “Router”, “Switch”)} \\
\ell_E &: E \to \{\text{Inter}, \text{Intra}, \text{Intranode}, \ldots\} && \text{(edge kind)} \\
\pi_V &: V \to (\text{String} \rightharpoonup \text{JSON}) && \text{(property maps)} \\
\pi_E &: E \to (\text{String} \rightharpoonup \text{JSON}) && \text{(edge property maps)} \\
L &\quad \text{(ordered set of named layers)} \\
\lambda &: V \to L \cup \{\perp\} && \text{(optional layer assignment)}
\end{aligned}
$$

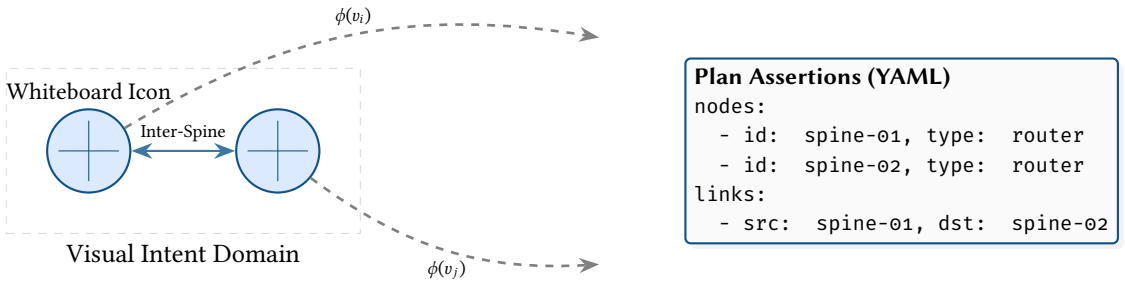


Figure 2. The W2P Isomorphism: Mapping Visual Intent to Formal Blueprint.

Nodes are assigned integer IDs that are globally unique within a topology instance. NTE uses `i32` as the external ID type. IDs are chosen by the caller; NTE does not auto-assign them (though helpers exist for auto-increment).

2.2 Node Types

NTE recognises two top-level vertex categories:

Node A network *device* — a router, switch, server, firewall, optical transponder, etc. Nodes carry a string `node_type` and an optional layer membership. Property data is stored as an arbitrary JSON object, populated by `ank_pydantic` from Pydantic model fields.

Endpoint A connection point on a node — a physical or logical port, interface, or sub-interface. Endpoints are owned by exactly one node via an Intra edge. Like nodes, they carry a type label (e.g. "Endpoint", "Interface") and a property map.

A **LogicalNode** is a specialised node that *always* belongs to a layer and represents protocol-level or virtual topology objects (e.g. a BGP session state machine, a VLAN domain). Logical nodes are excluded from most query results by default.

> **Terminology note**
>
> NTE uses "node" specifically for *network devices* and "endpoint" for their connection points. This differs from graph-theory usage where "node" means any vertex. In this document, "vertex" is used when the graph-theoretic meaning is intended.

2.3 Edge Kinds

NTE defines three principal external edge kinds. Internally, the engine also uses several bookkeeping edge kinds (`Parent`, `Child`, `Root`, `Origin`) that are invisible to Python callers by default.

2.3.1 Inter (Connectivity)

An **Inter** edge connects two endpoints. It models a physical or logical link between two ports on different devices. Inter edges are bidirectional by convention: NTE stores them as a pair of directed edges. The Python method is `add_inter_edges`.

2.3.2 Intra (Structural / Ownership)

An **Intra** edge runs *from* an endpoint *to* the node that owns it. It expresses the "this port belongs to this device" relationship. A node may own any number of endpoints; each endpoint has exactly one owning node. The Python method is `add_intra_edges`.

2.3.3 Intranode

An **Intranode** edge connects two nodes within the same logical chassis. This is used to model devices that consist of multiple line cards or virtual routing instances that share a common management plane but appear as separate nodes in the topology. The Python method is `add_intranode_edges`.

2.3.4 Device Shortcut (DeviceLink)

When an Inter edge is created between two endpoints, NTE automatically adds a **DeviceLink** shortcut edge between the owning devices. This avoids the three-hop traversal (*DeviceA* → *PortA* → *PortB* → *DeviceB*) in the common case where callers only want device-to-device connectivity. The shortcut is maintained in `device_shortcuts: HashMap<(i32,i32), Vec<EdgeIndex>>` inside `NteGraph`.

2.4 The Three-Hop Connectivity Pattern

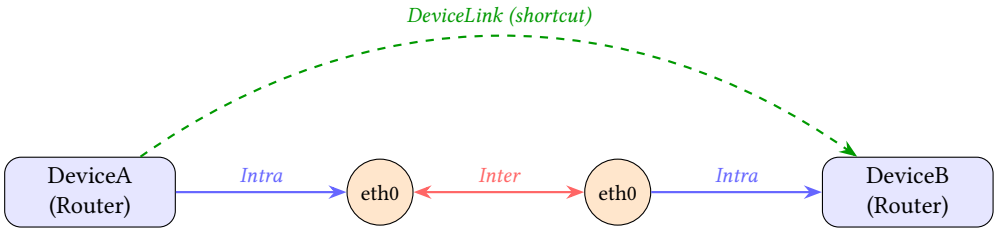


Figure 3. The three-hop connectivity pattern and the automatic DeviceLink shortcut. Intra edges run *from* endpoint to device; Inter edges are bidirectional.

Figure 3 shows the canonical pattern for inter-device connectivity. The DeviceLink shortcut means that "which devices are directly connected?" queries do not need to traverse six vertices; they can follow a single shortcut edge. However, if a query requires port-level detail (e.g. "which port on DeviceA connects to DeviceB?"), the full three-hop path is available in the graph.

> **Abstraction Leak**
>
> The necessity of the `DeviceLink` shortcut highlights an abstraction leak: the three-hop reality of network ports complicates basic queries. Future iterations of the NTE-QL planner aim to automatically collapse endpoint traversals when querying at the device level, removing the cognitive burden from the user.

2.5 Layer Hierarchy

Layers partition the topology into planes of abstraction. Common examples include a `physical` layer (hardware links), an `ip` layer (IP adjacencies), a `bgp` layer (BGP sessions), and an `mpls` layer. Layers are ordered and can declare dependencies: the BGP layer depends on the IP layer, which depends on physical.

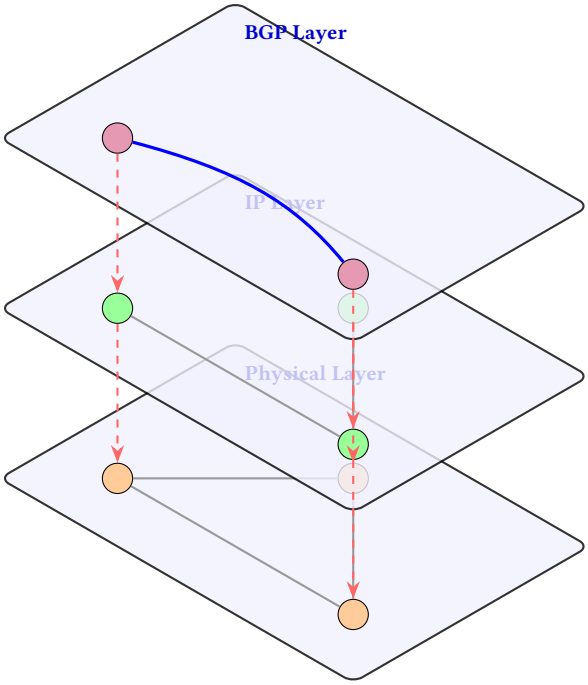


Figure 4. The 3D layered topology model. Logical overlays (BGP, IP) project down onto their physical underlays via structural dependencies.

Layers are stored in a `LayerDependencyGraph` (a DAG of layer names). NTE validates at edge-creation time that nodes on incompatible layers are not mixed (raising `LayerMismatchError` if so). A special `base` layer exists for cross-cutting nodes that appear in all layers.

2.6 Multi-Layer Topology Invariants

A fundamental strength of the NTE engine is its ability to maintain structural invariants across the vertical stack. Most network models treat physical and logical layers as isolated graphs; NTE formalises their relationship through **Ancestor/Descendant** mappings.

: Cross-Layer Structural Integrity

Every logical node n_L in layer L_i must have a structural mapping $\mathcal{M} : n_L \to n_P$ where n_P is an underlay node in layer $L_j (j < i)$. If a physical node is removed, all its logical descendants must be either re-parented or marked as "orphaned" to maintain referential integrity.

NTE provides optimized primitives for "tunneling" through the stack:

- `get_ancestor_in_layer(node_id, layer)`: Efficiently walks the hierarchy to find the physical hardware (e.g., Switch Port) supporting a high-level session (e.g., BGP Peer).
- `get_descendant_in_layer(node_id, layer)`: Projects a physical failure up the stack to identify all impacted logical services.

This cross-layer awareness ensures that queries like *"Find the physical switch port supporting this BGP session"* are resolved in $O(H)$ time, where H is the height of the layer stack, typically ≤ 5.

2.7 State Reconciliation: Intent vs. Observed

A critical requirement for modern network automation is the ability to distinguish between the **Intent** (what the designer wants) and the **Observed State** (what the telemetry reports). NTE treats these as two distinct but isomorphic representations of the same topology.

: Truth Isomorphism

Every node and edge in NTE carries a `source` attribute, either `intent` or `observed`. The engine provides built-in `reconcile()` primitives that compute the graph-structural and property-relational "Drift" between these two worlds.

By storing both Intent and Observed state in the same hybrid graph-relational engine, NTE enables sophisticated drift analysis:

- **Structural Drift**: Detecting missing links or unexpected hardware that exists in the physical world but not in the blueprint.
- **Property Drift**: Identifying configuration mismatches where the operational speed or MTU of an interface disagrees with the design.

The result of a reconciliation operation is a `TopologyDelta` that can be used to either "push" the intent to the network or "pull" the observed state back into the blueprint to reflect reality.

2.8 Property Maps

Both nodes and edges carry arbitrary key-value properties stored as JSON objects. Properties are serialised from Pydantic model instances by `ank_pydantic` and stored in Polars DataFrames as individual columns (one column per field name). The Polars representation enables fast columnar scans, type-aware comparisons, and SIMD acceleration.

Warning

Property columns are created lazily as new field names appear. This means queries against fields that have never been set on any node return an empty result rather than an error. Check column existence before relying on absent-field semantics.

2.9 Workspace Layout

NTE is an 18-crate Cargo workspace. The crates and their roles:

Crate	Layer	Purpose
`src/` (root)	Bindings	PyO3 Python extension (`lib.rs`, `topology.rs`, `query.rs`, `transaction.rs`)
`nte-core`	Core	Core topology with domain bindings
`nte-domain`	Core	Pure domain types (no dependencies)
`nte-graph`	Core	`petgraph` wrapper, ID mapping, CSR, GPU kernels
`nte-topology`	Core	Topology struct, transactions, snapshots, diffing, shadow
`nte-query`	Query	Executor, planner, matcher, profiler, visualisation, algorithm plugins
`nte-cli`	Tooling	NTE-QL REPL, policy validator, TUI dashboard
`nte-lsp`	Tooling	Language Server Protocol for NTE-QL editing
`nte-policy`	Policy	Starlark engine, DeltaNet validator
`nte-backend`	Storage	Abstract `TopologyBackend` trait
`nte-datastore`	Storage	Feature-flag crate selecting the active backend
`nte-datastore-core`	Storage	Shared backend types
`nte-datastore-polars`	Storage	Polars Arrow-backed DataFrames (default)
`nte-datastore-duckdb`	Storage	DuckDB embedded OLAP (experimental)
`nte-datastore-lite`	Storage	Lightweight in-memory store (testing)
`nte-server`	Server	Axum HTTP/WebSocket, Raft consensus, diagnostics
`nte-monte-carlo`	Optional	Monte Carlo simulation
`ladybug_backend`	Experimental	KuzuDB graph-native backend exploration

Figure 5 shows how the crates relate. Data flows downward from the consumer frontends through the PyO3 binding layer into the core engine. The query planner sits between the consumer and the dual-write store, deciding whether to consult the graph or the columnar store first.

3 Policy Engine

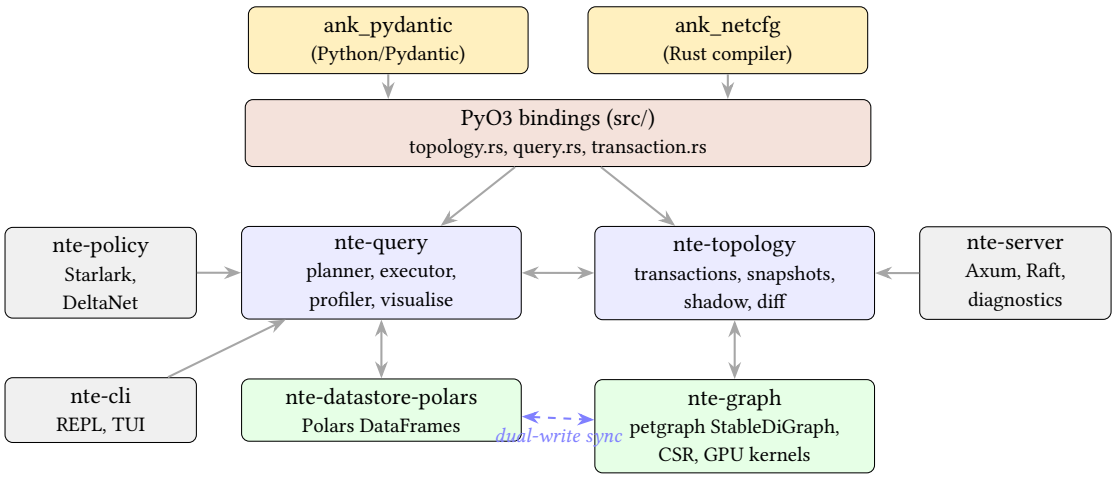


Figure 5. Crate architecture and data flow. Solid arrows show dependency/call direction; the dashed arrow represents the dual-write consistency protocol between the columnar and graph stores.

3.1 Design Rationale

Network topologies must satisfy design-rule constraints: every edge device must have at least two uplinks, no two devices in the same rack should be connected to the same spine, all transit links must have ≥ 100 Gbps speed. Encoding these constraints as ad-hoc Python checks is fragile and scattered across the codebase.

NTE externalises design-rule contracts into a declarative policy layer [3], separating *what the constraint is* from *when it is evaluated*. Policies can be evaluated on demand, before a mutation (what-if), or as a continuous validation step in a CI pipeline.

: Shadow Topology Isolation

NTE leverages its copy-on-write (CoW) architecture to support "What-If" analysis via Shadow Topologies. A mutation can be applied to an ephemeral child snapshot, validated against the full policy engine, and discarded without ever modifying the primary production state. This provides Correctness-by-Construction for network-wide changes.

3.2 Policy File Format

Policies are defined in YAML files with the following schema:

Listing 1. Policy file format (YAML)

```yaml
version: 1
policies:
  - id: "P001"
    category: "attribute"     # or "structural"
    severity: "ERROR"         # or "WARNING"
    expr: "device.speed_gbps >= 100"
    message: "All devices must support 100G"
    repair_hints:
      - "Upgrade device hardware"
      - "Replace with 100G-capable model"

  - id: "P002"
    category: "structural"
    severity: "ERROR"
    expr: "transit_exit_count >= 2"
    message: "Every PoP must have at least 2 transit exits"
    repair_hints:
      - "Add a second transit provider"
```

Two policy categories are supported:

`attribute` Evaluated per node/edge. The expression is checked against each entity's property map independently.

`structural` Evaluated once against global topology metrics (counts, ratios, aggregate properties).

3.3 The Starlark Policy Engine

For policies requiring programmatic logic beyond simple comparisons, NTE includes a Starlark-based policy engine. Starlark is a Python-like scripting language designed for safe, deterministic execution in embedded environments. It was chosen because:

- **Sandboxed**: no filesystem access, no network access, no mutable global state.
- **Deterministic**: no random number generation, no timing functions.
- **Familiar**: syntax is a subset of Python, comfortable for network engineers.
- **Fast**: scripts are parsed and compiled to bytecode once, then executed against each context.

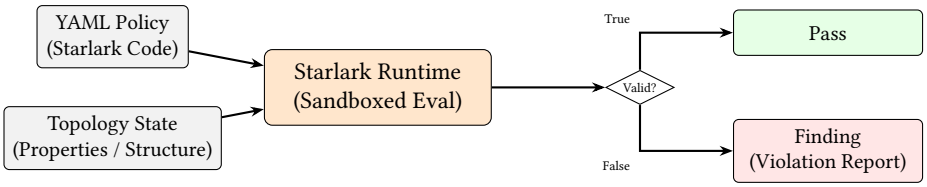


Figure 6. Starlark Policy Evaluation. The sandboxed runtime safely evaluates Python-like policy code against the topology state to generate deterministic compliance findings.

Listing 2. Starlark policy example: uplink redundancy

```
# Policy: every leaf switch must have at least 2 uplinks
def validate_leaf_uplinks(node_id, uplink_count, required):
    if uplink_count < required:
        fail("Node {} has only {} uplinks, need {}".format(
            node_id, uplink_count, required
        ))
    return True

# Host function bound from Rust: check_uplinks(node_id, required)
# Returns True if the node has >= required uplinks
result = check_uplinks("leaf-01", 2)
result  # Starlark scripts return the value of their last expression
```

Host functions are registered via the `#[starlark_module]` macro in Rust:

Listing 3. Registering host functions in the Starlark engine

```
#[starlark_module]
pub fn nte_graph_funcs(builder: &mut GlobalsBuilder) {
    fn check_uplinks(node_id: &str, required: i32) -> anyhow::Result<bool> {
        // Performs a petgraph degree lookup against the current Topology context.
        // Returns true if the node has >= required uplinks.
        ...
    }
}
```

3.4 Policy Evaluation (CEL & Starlark)

The `PolicyEngine` struct pre-compiles all policy expressions at load time. To balance performance and expressivity, NTE employs a tiered execution model:

CEL (Common Expression Language) Used for high-performance attribute and structural checks. CEL is a C++-like declarative language designed for safe, fast evaluation. It is ideal for simple property comparisons and existence checks.

Starlark Used for complex procedural logic that requires loops, conditionals, or custom functions (e.g., recursive path validation).

At evaluation time, the engine performs the following steps:

1. **Global Context Injection**: Structural policies are evaluated once against a global `RuntimeContext` containing aggregate topology metrics (e.g., total node count, layer distribution).
2. **Subject Iteration**: Attribute policies are evaluated for every matching subject (node or edge). The engine binds `this` (the entity ID) and `attrs` (the property map) into the evaluation scope.
3. **Deterministic Reporting**: Each violation produces a stable `Finding` report. These reports are deterministic and can be exported as JSON (v1 schema) for automated reconciliation.

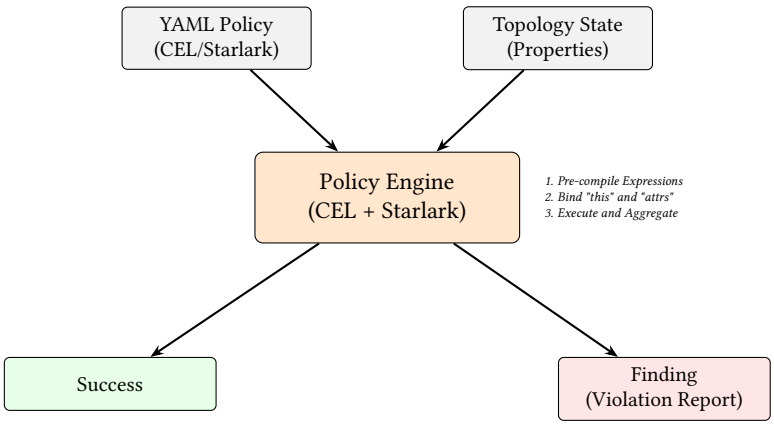


Figure 7. Policy Evaluation Pipeline. The engine pre-compiles declarative CEL expressions and procedural Starlark logic, executing them against a unified topology context to generate structured findings.

3.5 Shadow Topologies and DeltaNet Validation

NTE supports *what-if* analysis via `ShadowTopology`: a clone of the live topology onto which a proposed `TopologyDelta` is applied without affecting the real state.

Warning

The Snapshot Memory Tax: While the Polars DataFrames utilize Arrow CoW buffers (making property cloning effectively $O(1)$), `petgraph::StableDiGraph` is backed by standard Rust `Vecs`. Creating a shadow topology requires an $O(|V|+|E|)$ deep memory copy of the entire structural graph. For topologies exceeding 100,000 nodes, creating a shadow (or starting a transaction) will incur a millisecond-level CPU stall.

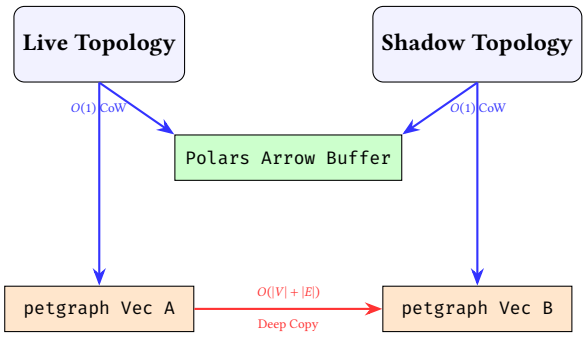


Figure 8. The Hybrid Snapshot Reality. While property DataFrames share memory instantly via CoW pointers, the underlying petgraph adjacency lists must be physically deep-copied into new memory allocations.

3.6 DeltaNet: Incremental Policy Validation

The `DeltaNetValidator` (in `nte-policy`) provides an optimized alternative to global topology validation. Rather than re-evaluating every policy against the entire graph, DeltaNet uses an **Impact Radius** algorithm to identify the minimal set of nodes affected by a mutation.

: Impact-Aware Validation

For any mutation Δ, the validator computes the affected subgraph $\mathcal{G}_\Delta \subseteq \mathcal{G}$ using a bounded BFS traversal of depth k (where k is the maximum dependency depth of the active policies). Only nodes within $\mathcal{G}_\Delta$ are re-validated, reducing validation complexity from $O(V)$ to $O(V_\Delta)$.

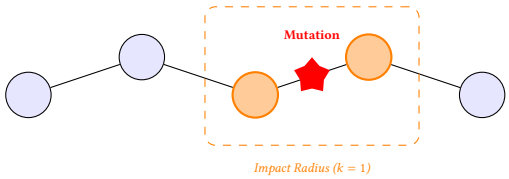


Figure 9. Incremental Validation via Impact Radius. By identifying the nodes physically and logically adjacent to a mutation, NTE avoids the $O(V)$ cost of global policy re-evaluation.

This incremental approach is essential for large-scale production networks where a single property update (e.g., a BGP weight change) should not trigger a multi-second validation sweep of 100,000 devices.

3.7 Worked Policy Examples

3.7.1 Transit Redundancy

Listing 4. Transit redundancy policy

```
# Every PoP must have at least 2 transit exit links
def check_transit_redundancy(pop_name, exit_count):
    if exit_count < 2:
        fail("PoP {} has only {} transit exits (minimum 2)".format(
            pop_name, exit_count
        ))
    return True

check_transit_redundancy(pop_name, transit_exit_count(pop_name))
```

3.7.2 Peer Diversity

Listing 5. CEL attribute policy: peer diversity

```
# P003: No two BGP sessions should share the same peer AS (from a given router)
# Expressed as a structural policy on the aggregated adjacency matrix
version: 1
policies:
  - id: "P003"
    category: "structural"
    severity: "WARNING"
    expr: "max_sessions_per_peer_as <= 1"
    message: "Router has multiple BGP sessions to the same peer AS"
    repair_hints:
      - "Use route reflectors or consolidate peering"
```

3.7.3 No Single Point of Failure

Listing 6. No single point of failure policy

```
# All routers in the critical path must have >= 2 uplinks
def check_no_spof(critical_nodes):
    for node_id in critical_nodes:
        uplinks = check_uplinks(node_id, 2)
        if not uplinks:
            fail("Critical node {} is a single point of failure".format(node_id))
    return True

# critical_path_nodes() is a host function that returns the set of
# articulation points in the physical topology
check_no_spof(critical_path_nodes())
```

Part II

Engine Implementation (Deep Dive)

4 Storage Architecture

Storage Architecture at a Glance

Why: Combining property filtering and graph traversal efficiently requires specialized structures.
What: A dual-write system. Columnar DataFrames for fast property scans, and adjacency lists (Petgraph) for structural paths.

4.1 Rationale for Dual-Write

A graph database excels at traversal. A columnar store excels at predicate evaluation on properties. For network topology queries, you almost always need both. A query like "find all routers in the SYD PoP that have a BGP session to a peer with AS 65000" has two parts:

- *Property filter*: `pop == "SYD"`, best served by a columnar scan.
- *Structural filter*: connected via a BGP-layer edge to a peer with `as_number == 65000`, best served by graph traversal.

NTE's dual-write approach keeps both representations in sync and lets the query planner pick the optimal execution order.

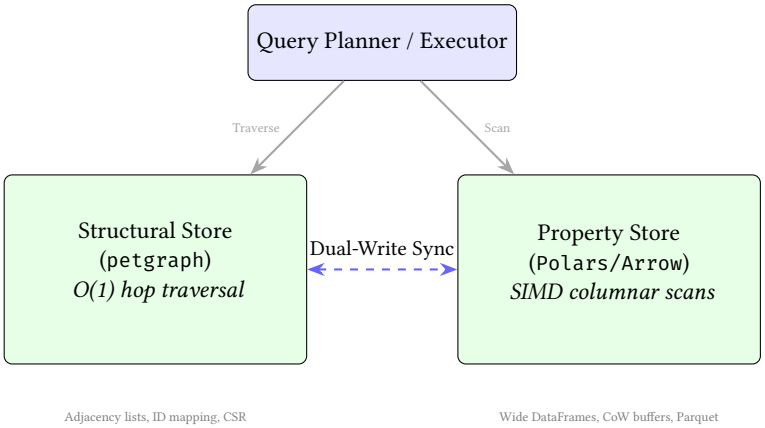


Figure 10. Dual-write architecture synergy. Structural queries target the graph engine, while property-heavy predicates target the columnar store.

4.2 petgraph StableDiGraph

The graph structure is stored in a `petgraph::StableDiGraph<GraphNode, GraphEdge>` [4]. `StableDiGraph` was chosen over the non-stable variant because:

- Node and edge indices remain valid after deletions. This is essential because NTE's ID mapping (`HashMap<i32, usize>` in both directions) must not be invalidated by unrelated mutations.
- It allows $O(1)$ lookup of node weights by `NodeIndex`, which is the primary graph-layer operation.

The `NteGraph` struct wraps `StableDiGraph` and adds:

- **Bidirectional ID mapping**: `index_map: HashMap<i32, usize>` (external → internal) and `reverse_index_map: HashMap<usize, i32>` (internal → external). All Python-facing APIs use external `i32` IDs.
- **Endpoint / node ID sets**: `HashSet<i32>` for $O(1)$ "is this ID an endpoint?" checks.
- **Layer dependency graph**: `LayerDependencyGraph` for topological ordering and layer-consistency checks.
- **Device shortcut map**: `HashMap<(i32,i32), Vec<EdgeIndex»` for fast device-to-device lookups.
- **CSR cache**: a thread-safe, lazily-computed Compressed Sparse Row representation for high-performance batch traversal (described in Section 8).

4.3 Polars Columnar Store

Node and edge properties are stored in Polars [5] DataFrames — one DataFrame per entity class (nodes, endpoints, links). Polars uses the Apache Arrow columnar format internally, which gives:

- SIMD-accelerated predicate evaluation (AVX2/AVX-512 on x86-64).
- Copy-on-Write (CoW) semantics for zero-copy snapshot cloning.
- Lazy evaluation via `LazyFrame` for filter push-down.

Properties are stored in wide columnar format: each unique property name becomes a column in the DataFrame. This is a deliberate trade-off. A narrow schema (one row per property) would be more flexible but would require an expensive "pivot" before columnar predicates could be applied. The wide schema means NTE scans efficiently but requires that all instances of a given node type have compatible types for a given field name.

: Sparse Columnar Compression

A hyper-scale topology generates highly heterogeneous schema (e.g., a firewall has `fw_rules`, a router has `bgp_asn`). The engine must aggressively map all unique JSON keys to a wide columnar schema. Storage explosion is avoided via Apache Arrow's validity bitmaps, ensuring that null values consume less than 1 bit per row while remaining perfectly aligned for SIMD vectorization.

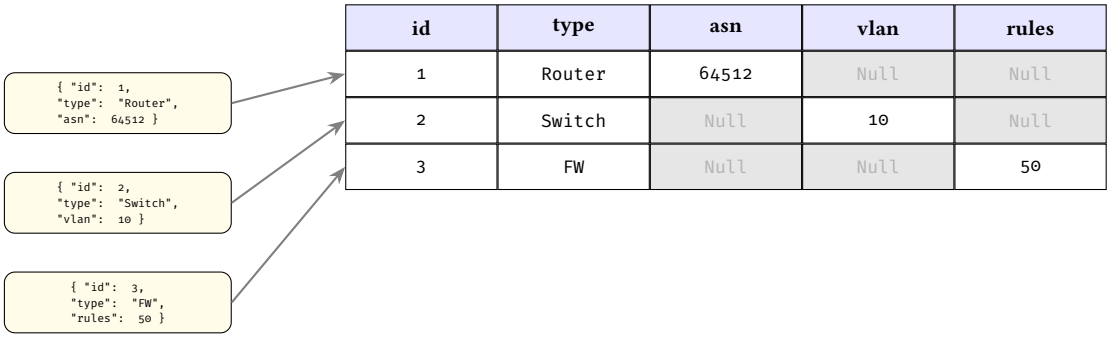


Figure 11. The Wide-Column Sparsity Matrix. Heterogeneous JSON properties are flattened into a single wide DataFrame. Arrow's validity bitmaps efficiently compress the resulting `Null` regions.

4.4 Memory Fragmentation and Defragmentation

A side effect of using `StableDiGraph` for its index stability is the creation of "tombstones" (empty slots in internal arrays) when nodes or edges are deleted. In mutation-heavy workloads, such as iterative network synthesis or large-scale failure-model simulations, these holes destroy CPU cache locality and increase memory footprint.

: Dynamic Memory Packing

To maintain hyper-scale performance, the engine must decouple *logical index stability* from *physical memory density*. NTE provides an atomic `pack()` operation that rebuilds the graph structure into a contiguous layout while preserving the isomorphism with the relational property store.

The `pack()` operation executes in $O(V + E)$ time by:

1. Allocating a new, dense `StableDiGraph` with exactly the capacity needed.
2. Mapping old node indices to new, contiguous indices.
3. Rewriting all internal state (ID maps, device shortcut caches, and edge indices embedded in weights) to reflect the new physical layout.

This operation is exposed to the Python layer as `defragment_memory()`, allowing long-running automation workers to periodically restore cache efficiency without losing data integrity.

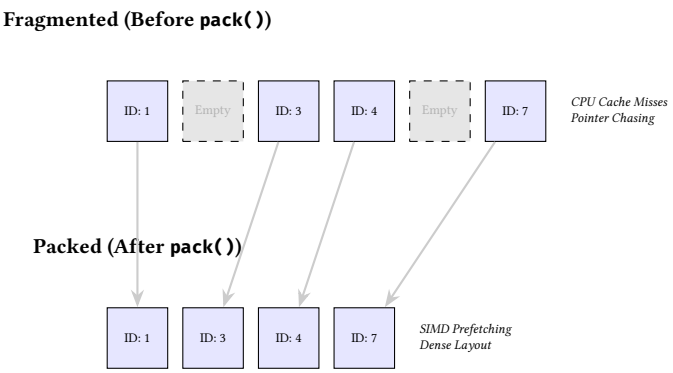


Figure 12. Memory Defragmentation (Graph Packing). The `pack()` operation removes tombstone slots from the internal adjacency arrays, restoring linear CPU cache locality and reducing memory overhead.

SIMD Vectorization. Unlike row-based graph databases that suffer from high cache-miss rates due to "pointer chasing," NTE leverages the linear memory layout of Polars to perform SIMD-accelerated predicate evaluation. By loading property columns into 512-bit registers (AVX-512), the engine can evaluate up to 16 integer comparisons in a single CPU clock cycle.

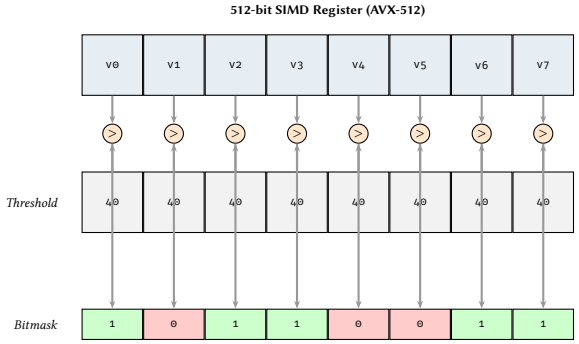


Figure 13. Hardware-Level Predicate Evaluation. A single AVX-512 instruction evaluates multiple property values in parallel, producing a bitmask that drives the subsequent graph traversal filter.

4.5 Dual-Write Consistency Protocol

Every mutation method in the `Topology` struct applies changes to both the graph and the DataFrame store before returning. The protocol is:

Algorithm 1 Dual-write mutation protocol

1: Acquire write lock on topology
2: Validate inputs (type checks, layer compatibility, ID uniqueness)
3: Apply mutation to `NteGraph` (structural side)
4: Apply corresponding mutation to `DataFrameStore` (property side)
5: If either step fails, rollback and propagate error
6: Emit `TopologyEvent` to ring buffer
7: Invalidate CSR cache
8: Release write lock

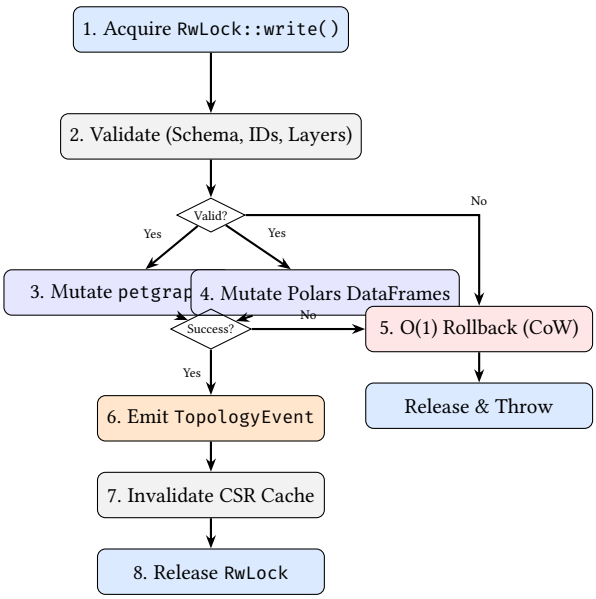


Figure 14. The dual-write transaction lifecycle. Copy-on-Write semantics allow $O(1)$ rollback if either the graph or the DataFrame mutation fails.

> **Note**
>
> NTE implements copy-on-write transaction semantics: if step 4 fails, the graph mutation in step 3 is rolled back via a compensating operation and the DataFrame store is restored from its pre-mutation Arrow CoW clone in O(1). This ensures atomicity within a single process. Durability is provided via the **Write-Ahead Log (WAL)**: every committed mutation is appended to a persistent journal before the in-memory state is updated. Following a crash, the engine can reconstruct the exact topological state by replaying the WAL events (Section 7).

4.6 Foreign Function Interface & Zero-Copy Boundaries

The engine exposes a foreign function interface (FFI) primarily via PyO3, which allows NTE to embed directly into a Python process. This approach is significantly faster than external RPC calls to a standalone graph database. To maximize concurrency and performance, NTE implements several boundary optimisations:

- **GIL Release**: The `Topology` struct is protected by a `RwLock`. Query operations acquire a read lock; mutations acquire a write lock. Crucially, the PyO3 binding layer releases the Python Global Interpreter Lock (GIL) before acquiring these Rust locks. This allows concurrent Python threads to execute native code or I/O while another thread holds a write lock in Rust.
- **Zero-Copy PyArrow**: When the `arrow-ids` feature is enabled, Polars DataFrames are shared with Python using the Arrow C Data Interface. This eliminates serialisation overhead, allowing Python data science tools to interact with NTE's columnar store seamlessly and in zero-copy fashion.

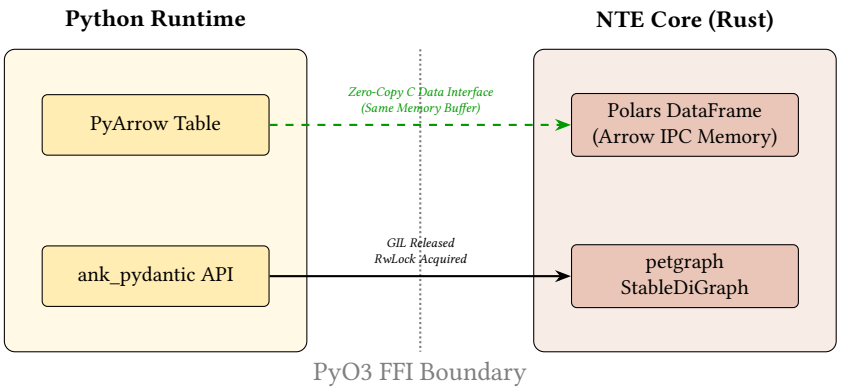


Figure 15. The FFI Zero-Copy Boundary. Unlike traditional RPC-based graph databases, NTE embeds directly in the Python process. The Arrow C Data Interface allows Python tools to read the Rust memory buffers without any serialization overhead.

4.7 EventStore Ring Buffer

Every mutation emits a `TopologyEvent` variant into an in-memory ring buffer (a `VecDeque` managed by the `EventStore`). Events carry:

- A monotonically increasing `EventSequence` number (atomic `u64`).
- A UTC timestamp (`chrono::DateTime<Utc>`).
- Event-specific payload (node IDs, edge pairs, changed fields, etc.).

Events are serialisable to JSON (via `serde_json`) for external consumption. The ring buffer has a configurable capacity; once full, the oldest events are evicted.

4.8 Pluggable Backends

The storage layer is factored into three crates:

Crate	Description	Status
`nte-datastore-polars`	Polars Arrow-backed DataFrames	Default, production
`nte-datastore-duckdb`	DuckDB embedded OLAP database	Optional, experimental
`nte-datastore-lite`	Lightweight in-memory store	Development / testing

The `nte-datastore-core` crate defines the abstract `TopologyBackend` trait that all backends implement. Selecting a backend is a compile-time feature flag in `nte-datastore`.

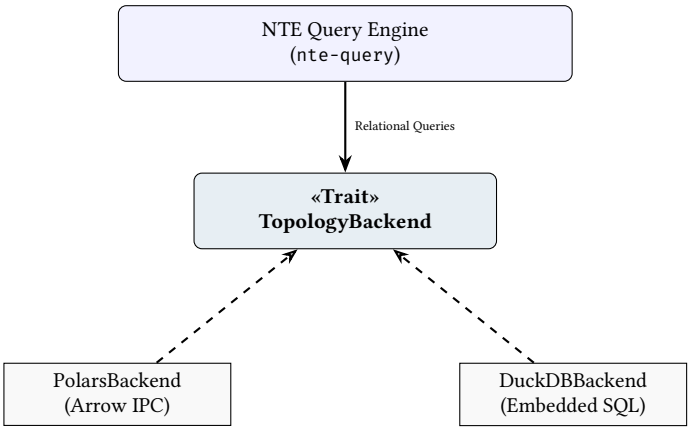


Figure 16. Trait-Based Abstraction. The core query engine interacts only with the `TopologyBackend` interface, allowing the physical columnar storage (Polars vs. DuckDB) to be swapped at compile time.

5 Query System

> **Query System at a Glance**
>
> **Why**: Naive graph traversal is slow when properties are dense and selective.
> **What**: A filter-first planner that drastically reduces datasets via parallel column scans before invoking structural graph traversal.

5.1 Design Goals

NTE's query system was designed around three principles:

1. **Filter-first execution**: property predicates should run before graph traversal, not after, to eliminate candidates early.
2. **Composable, typed query objects**: queries should be buildable programmatically from Python, not only via string parsing, so that tooling can construct, inspect, and serialise query plans.
3. **Honest cost estimation**: the planner should rank predicates by selectivity rather than always applying them left-to-right.

5.2 Query Representations

The `nte-query` crate exposes two parallel query representations:

`QuerySpec` A flat set of node-level filters: type, layer, kind, ID set, field equality, and expression filters. Used for "give me all nodes matching these conditions" queries. Exposed as `QuerySpec` in Python.

`QueryPlan` / `PatternNode` A structural graph pattern AST. Describes a subgraph shape to match (e.g. a router connected to a switch via an Inter edge). Used for multi-hop traversal queries. Exposed as `QueryPlan` and `PatternNode` in Python.

: Predicate Pushdown

In the hybrid query planner, property filters and type assertions must be pushed down to the relational store (Polars) before any structural traversal begins. This reduces the candidate node set via SIMD vectorization, effectively converting a complex $O(V + E)$ graph search into a constrained $O(K)$ bounded walk, where $K \ll V$.

5.3 Query Lowering and Execution

Before execution, high-level NTE-QLstrings or Python `QueryPlan` objects are "lowered" into a hybrid execution pipeline. This process decomposes the query into its relational and structural components.

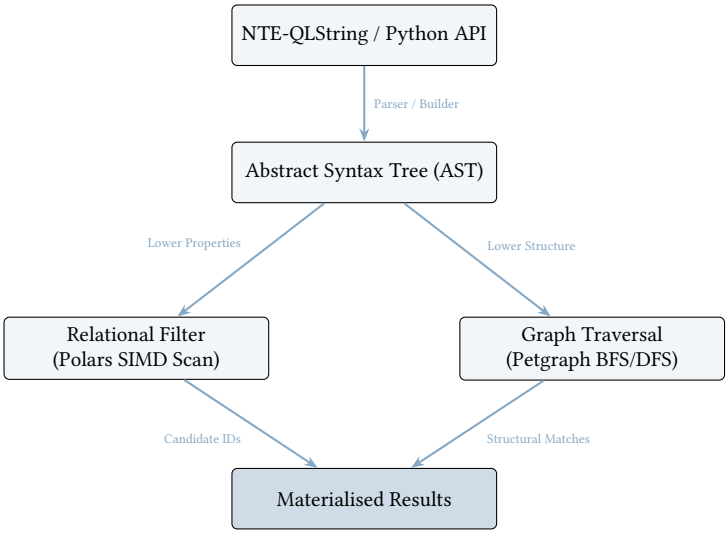


Figure 17. The Query Lowering Pipeline. High-level intent is decomposed into parallel relational and structural execution tasks.

The planner uses a **Cost-Based Lowering** strategy: if a query contains highly selective property filters (e.g., a specific `hostname`), the relational scan is executed first to prune the graph search space. If the query is purely structural, it is lowered directly to the petgraph engine.

5.4 Query Complexity: SIMD Pruning vs. Adjacency Traversal

The performance advantage of NTE's hybrid approach can be quantified using Big-O analysis. Consider a property-heavy structural query (e.g., "Find all Routers where pop='SYD' and follow their 1-hop links").

Naive Traversal cost $O(V+E)$. In a row-based graph store, we must visit every node V to check the property, then traverse its adjacency list E. This is dominated by pointer chasing and cache misses.

NTE Filter-First cost $O(\frac{V}{K} + E_{sub})$. where K is the SIMD vectorization factor ($K = 16$ for AVX-512) and E_{sub} is the set of edges connected only to the matching nodes.

Since $E_{sub} \ll E$ in selective queries, the query time is reduced by approximately an order of magnitude. The relational scan $O(V/K)$ is a linear scan of contiguous memory, maximizing cache locality and CPU throughput.

5.5 QuerySpec: The Relational Path

The `QueryExecutor` in `nte-query` compiles a `QuerySpec` into an ordered `FilterPlan` — a sequence of `FilterOp` variants sorted by `FilterRank`:

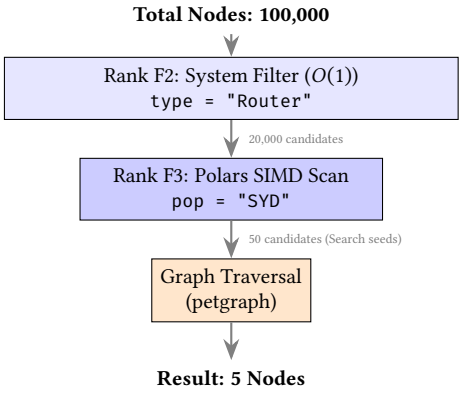


Figure 18. The Cardinality Funnel (Filter-First Execution). Cheap columnar operations aggressively prune candidates, ensuring the more expensive graph traversal engine only visits a tiny fraction of the topology.

Execution proceeds left-to-right. Each filter receives the candidate set from the previous filter and returns a (potentially smaller) subset. The Polars LazyFrame pipeline is constructed in-order and materialised with a single `collect()` call, benefiting from Polars' internal optimiser.

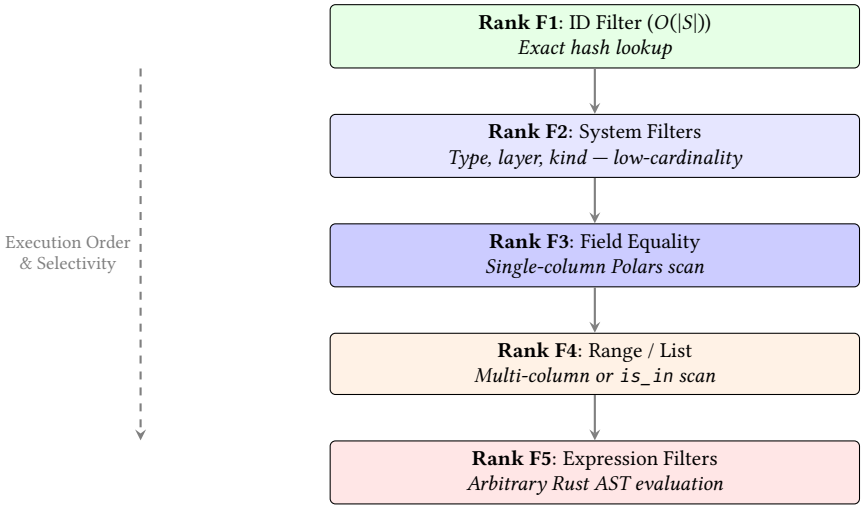


Figure 19. Query Planner Execution Ranks. Predicates are sorted and executed from cheapest (F1) to most expensive (F5) to continuously reduce the candidate set.

Filter plan example

Query: "All routers in the SYD PoP with AS number 64512"

1. F2: type = "Router" → scan `node_types` DataFrame
2. F3: `pop == "SYD"` → filter `data_pop` column
3. F3: `as_number == 64512` → filter `data_as_number` column

Polars executes steps 2 and 3 as a single lazy pipeline after step 1 narrows the candidate set.

5.6 Pattern Matching

The `PatternNode` AST describes the shape of a subgraph. Supported patterns:

`PatternNode::node(type)` Match a single vertex of the given type.

`PatternNode::binding(name, type)` Match a vertex and bind it to a name for later reference.

`PatternNode::chain(steps)` A sequence of alternating node-steps and edge-steps.

`PatternNode::union(patterns)` Union of multiple patterns.

Variable-length paths `HopSpec::exact(n)`, `HopSpec::range(min,max)`, or `HopSpec::any()`.

The pattern matcher performs a depth-first search over the graph, pruning branches when a node fails the pattern constraint. The maximum number of matches is capped at `QueryPlan::DEFAULT_MAX_MATCHES` (10,000) by default to prevent runaway traversals; the result carries a `truncated` flag when this cap is reached.

5.7 Selectivity Ranking in Detail

The five filter ranks correspond to the following `FilterOp` variants:

Listing 7. FilterRank enum from executor.rs

```
enum FilterRank {
    Id         = 0,  // HashSet<i32> lookup -- always first
    System     = 1,  // Type/layer/kind -- low-cardinality columns
    Equality   = 2,  // Single field equality
    Range      = 3,  // List / is_in / range
    Expression = 4,  // Arbitrary ExprNode AST -- always last
}
```

Field equality filters on list values are promoted from `Equality` to `Range` rank because they require a Polars `is_in` operation, which is costlier than a direct equality predicate.

5.8 Query Profiling

Setting `QueryHints{enable_profiling: true}` in a `QuerySpec` records timing data for each filter step. The profiler captures per-step candidate counts and wall-clock durations, enabling identification of expensive filters. The profile is returned alongside the result set.

5.9 Plan Cache

NTE maintains an LRU plan cache (capacity configurable, default 256 entries) for `QuerySpec` objects. The cache key is a `CacheKeyMaterial` struct containing sorted, canonicalised representations of all filter fields (to avoid cache misses from HashMap ordering non-determinism). Cache keys are computed lazily via `OnceLock` to avoid repeated sorting allocations.

> **Note**
>
> The plan cache caches compiled `FilterPlan` objects, not query results. Result caching (returning previously computed row sets) is available via `QueryHints{cache_results: true}` but is off by default, as topology mutations must invalidate the result cache.

5.10 Query API Decision Matrix

NTE provides three distinct query paradigms. To avoid fragmentation, callers should select their interface based on the following matrix:

Interface	Primary Audience	When to use
Programmatic (`QuerySpec`)	Integrators	Internal system-to-system queries where parameters are highly dynamic or built from ASTs.
Fluent Builder	Data Scientists	Interactive Python notebooks; chaining complex property filters.
NTE-QL (Cypher)	Network Engineers	Human-readable graph pattern matching, CLI/TUI usage, and external dashboarding.

5.11 NTE-QL String Interface

In addition to the programmatic query API, NTE supports a Cypher-inspired string query language called NTE-QL. Queries are parsed by a PEG grammar implemented with the `chumsky` parser combinator library and compiled into `QueryPlan` / `QuerySpec` objects before execution.

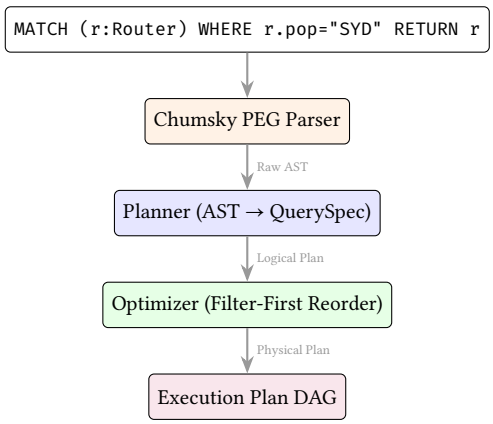


Figure 20. The NTE-QL compilation pipeline. String queries are parsed, converted into programmatic `QuerySpec` structures, and fed into the same optimizer as the Python APIs.

The grammar supports a variety of expressions. Here is a brief survey of supported query patterns:

Simple node scan

```
MATCH (r:Router)
WHERE r.pop = "SYD" AND r.as_number = 64512
RETURN r.hostname
```

Two-hop traversal: direct BGP peers

```
MATCH (r:Router)-[:Inter]->(s:Router)
WHERE r.hostname = "core-01"
RETURN r.hostname, s.hostname, s.as_number
```

Bounded-hop reachability: within 3 hops

```
MATCH (src:Router)-[*1..3]->(dst:Router)
WHERE src.hostname = "edge-01"
RETURN dst.hostname
```

Aggregation: node degree

```
MATCH (r:Router)-[:Inter]->(s:Router)
WHERE r.pop = "SYD"
RETURN r.hostname, COUNT(s) AS peer_count
ORDER BY peer_count DESC
```

NOT / exclusion: isolated routers

```
MATCH (r:Router)
WHERE NOT EXISTS {
    MATCH (r)-[:Inter]->(:Router)
}
RETURN r.hostname
```

Path finding: shortest path

```
MATCH p = shortestPath((src:Router)-[*]->(dst:Router))
WHERE src.hostname = "r1" AND dst.hostname = "r4"
RETURN [n IN nodes(p) | n.hostname] AS path
```

5.12 The ExprNode AST

Complex filter predicates are represented as an `ExprNode` abstract syntax tree. The full set of node types:

Variant	Description
`Field(name)`	Reference to a node/edge property
`BindingField(binding, name)`	Property access via a named binding
`Literal(value)`	Constant (null, bool, int, float, string, list)
`Not(expr)`	Logical negation
`Logical(op, left, right)`	AND / OR
`Comparison(op, left, right)`	$=, \neq, <, \leq, >, \geq$
`Arithmetic(op, left, right)`	$+, -, \times, \div$
`StringOp(op, field, pattern)`	contains, startsWith, endsWith, matches
`IsNull(expr)` / `IsNotNull`	Null checks
`IsIn(field, values)`	Membership test

5.13 Python Fluent Query Builder

In addition to the programmatic `QuerySpec` / `PatternNode` APIs and the NTE-QLstring interface, NTE provides a fluent Python query builder that does not require `ank_pydantic`. The builder is inspired by Polars' expression API and compiles down to the same `QuerySpec` objects used by the Rust executor.

Fluent query builder: node queries

```python
from ank_nte import Topology
from ank_nte.query import QueryNamespace, Expr

t = Topology()
# ... populate topology ...
q = QueryNamespace(t)

# Chainable node query
syd_routers = (
    q.nodes()
     .of_type("Router")
     .in_layer("physical")
     .filter(Expr.field("pop") == "SYD")
     .sort("hostname")
     .ids()
)

# Expression DSL supports operators
fast_ports = (
    q.nodes()
     .of_type("Endpoint")
     .filter(Expr.field("speed_gbps") > 40)
     .count()
)

# String operations
juniper_devices = (
    q.nodes()
     .of_type("Router")
     .filter(Expr.field("vendor").contains("Juniper"))
     .collect()
)
```

Fluent query builder: link queries

```python
# Link queries for edge filtering
transit_links = (
    q.links()
     .of_type("Inter")
```

```
    .in_layer("physical")
    .filter(Expr.field("speed_gbps") >= 100)
    .ids()
)

# Between-nodes filtering
r1_links = (
    q.links()
     .between(source_ids=[1], target_ids=[2])
     .collect()
)
```

The expression DSL supports the full `ExprNode` AST via Python operator overloading: `==`, `!=`, `>`, `>=`, `<`, `<=` for comparisons; `&` (and), `|` (or), `~` (not) for logic; `+`, `-`, `*`, `/` for arithmetic; and methods `.contains()`, `.startswith()`, `.endswith()`, `.matches()` for string operations. Terminal methods include `.ids()`, `.collect()`, `.count()`, `.exists()`, `.first()`, and `.one()`.

5.14 Query Plan Visualisation

NTE can render query execution plans as Mermaid flowchart diagrams. This is useful for understanding how the planner decomposes a query and for identifying optimisation opportunities.

Listing 8. Mermaid DAG generation (from nte-query/src/visualize.rs)

```rust
pub fn generate_mermaid(plan: &ExecutionPlan) -> String {
    // Generates a Mermaid flowchart DAG from the execution plan.
    // Nodes represent operations: NodeScan, EdgeScan, Filter,
    // Join, WcojJoin (worst-case optimal), SemiJoin.
    // Edges represent data flow between operations.
}
```

The visualisation is accessible in two ways:

1. **NTE-QL REPL**: `EXPLAIN VISUAL <query>` renders the plan inline.
2. **HTTP API**: `GET /explain?query=...` returns an HTML page with the rendered Mermaid diagram (via the `nte-server` diagnostics endpoint).

5.15 NTE-QL Interactive REPL

The `nte-cli` crate provides a standalone command-line interface with three modes:

`nte-cli repl` An interactive NTE-QL shell (built on `reedline`) that parses queries, compiles them to execution plans, and displays results in formatted tables (via `comfy_table`). The `EXPLAIN` prefix shows the compiled plan; `EXPLAIN VISUAL` renders the Mermaid DAG.

`nte-cli validate` A policy validation workflow: loads a YAML policy file and a topology archive, evaluates all policies, and renders a compliance report.

`nte-cli dashboard` A TUI dashboard (built on `ratatui` and `crossterm`) showing real-time memory usage, CPU load, graph statistics (node/edge counts), and a 30-entry history sparkline.

NTE-QL REPL session

```
nte> MATCH (r:Router) WHERE r.pop = "SYD" RETURN r.hostname
+----------+
| hostname |
+----------+
| r-syd-01 |
| r-syd-02 |
+----------+
2 rows returned

nte> EXPLAIN VISUAL MATCH (r:Router)-[:Inter]->(s:Router) RETURN r, s
-- Mermaid DAG rendered --
```

6 End-to-End Query Trace

To illustrate the filter-first query planner's coordination between the relational and graph stores, we trace a representative multi-stage query:

"Find all spine routers in the SYD PoP with AS 64512 that have a < 3 hop path to a firewall."

The query execution follows three logical phases, as shown in Figure 21.

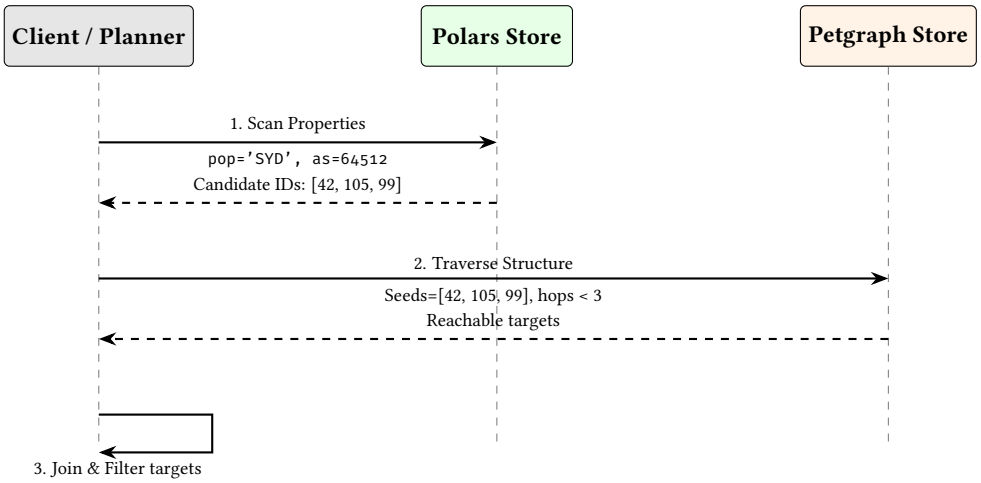


Figure 21. The Filter-First Pipeline. Relational scans prune the search space before expensive graph traversals are initiated.

Insight:
By using the Polars store to prune 99% of nodes based on properties (`pop`, `as_number`), the subsequent graph traversal only needs to visit a tiny fraction of the total adjacency list.

Operational Trace: What Actually Runs

In a real execution, Phase 1 emits a candidate ID vector from the columnar store. That vector becomes the seed set for Phase 2, where the traversal is bounded by a hop limit and early-terminated by predicate checks. Phase 3 performs a final join back to relational properties to render result rows and to produce an explainable plan trace for `EXPLAIN`.

7 Transactions

7.1 ACI (Durability via WAL) Transaction Model

Transaction Durability

NTE implements full **ACID** transactions (Atomicity, Consistency, Isolation, Durability). While structural mutations utilize copy-on-write (CoW) semantics for isolation, persistent durability is guaranteed by an optional **Write-Ahead Log (WAL)**. Every committed mutation is appended to a monotonically increasing journal on disk, ensuring that in-memory state can be recovered following a process crash.

Every mutation can be wrapped in a transaction scope that atomically commits or rolls back changes to both the graph and the DataFrame store.

The transaction model uses Arrow's reference-counted CoW buffers for $O(1)$ snapshot creation at transaction start and $O(1)$ rollback if the transaction is aborted. When WAL is enabled, the commit phase is a two-step process:

1. **Prepare**: The mutation is validated and prepared in-memory.
2. **Commit**: The event is flushed to the WAL on disk, then applied to the primary in-memory stores.

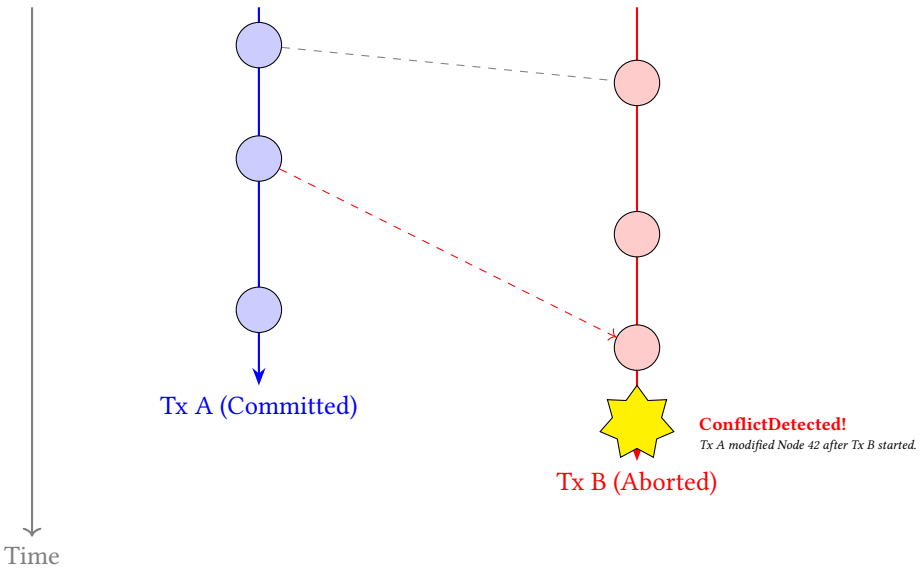


Figure 22. MVCC Transaction Timeline. Both transactions read from the same initial snapshot S_0. Because Tx A commits a mutation to Node 42 before Tx B attempts to mutate it, the Conflict Engine safely aborts Tx B to prevent a Write-Write conflict.

7.2 Isolation Levels

Three isolation levels are supported, matching standard database semantics:

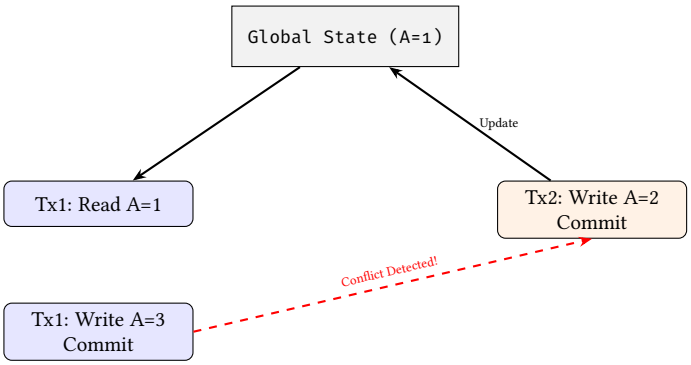


Figure 23. Serializable isolation conflict detection. Tx1 fails to commit because its read-set (A=1) was invalidated by Tx2's concurrent write.

Level	Behaviour
`ReadCommitted`	Sees only committed data; each read within the transaction sees the latest committed state.
`RepeatableRead`	Reads within the transaction see a consistent snapshot taken at transaction start.
`Serializable`	Full conflict detection: transactions that read/write overlapping data are detected and the later transaction is aborted.

7.3 Transaction Snapshots

Each transaction creates a `TransactionSnapshot` at begin time, using persistent data structures (`im::HashMap`) for structural sharing:

Listing 9. TransactionSnapshot (from nte-topology/src/topology/snapshot.rs)

```rust
pub struct TransactionSnapshot {
    pub adjacency: im::HashMap<NodeId, Vector<NodeId>>,
    pub node_metadata: im::HashMap<NodeId, NodeMetadata>,
    pub dataframes: Arc<DataFrameStore>,
}
```

The use of `im::HashMap` (an immutable persistent hash map) means that the snapshot shares structure with the live topology. Only modified entries are copied, giving near-constant memory overhead for transactions that touch a small fraction of the graph.

7.4 Python Transaction API

The `PyTransaction` class exposes transactions as a Python context manager:

Listing 10. Transaction API

```python
from ank_nte import Topology, IsolationLevel

t = Topology()
# ... populate topology ...

# Context manager: auto-rollback on exception
with t.begin_transaction(IsolationLevel.Serializable) as txn:
    t.add_nodes(ids=[100], node_types=["Router"], layer="physical",
                data=[{"hostname": "r-new"}])
    t.add_inter_edges(sources=[10], destinations=[100])
    txn.commit()  # explicit commit

# If an exception occurs before commit(), the transaction
# is automatically rolled back (both graph and DataFrame store).

# Manual rollback
with t.begin_transaction() as txn:
    t.remove_node_cascade(node_id=1)
    if not t.is_reachable(source=2, target=3):
        txn.rollback()  # revert: removal would partition the graph
    else:
        txn.commit()
```

Warning

NTE transactions are in-memory only. There is no write-ahead log (WAL): a process crash during a transaction leaves the in-memory state inconsistent. For durable crash consistency, use periodic disk snapshots or the Raft distributed deployment (Section 13).

7.5 Conflict Detection

When multiple transactions execute concurrently, NTE detects conflicts using a `VersionTracker` that records per-node and per-edge version numbers. At commit time, the tracker compares each transaction's read and write sets against committed versions:

Write–write conflicts (all isolation levels): if a node or edge was modified by another transaction after the current transaction began, a `ConflictError` is raised.

Read–write conflicts (Serializable only): if a node that was *read* by the current transaction was subsequently *written* by another committed transaction, a `ConflictError` is raised. Read-your-own-writes are excluded from this check.

Conflict errors carry the transaction ID, the list of conflicting node/edge IDs, and the conflict type (`WriteWrite` or `ReadWrite`), allowing callers to implement application-level retry or merge logic.

8 Persistence and Snapshots

: Zero-Copy Isolation

Snapshots must leverage Arrow's reference-counted, memory-mapped buffers to guarantee that historical state requires zero initial memory overhead. Cloning the property schema is an $O(1)$ pointer increment. This isolation underpins the engine's transactional rollbacks and "What-If" shadow topologies.

8.1 Named Snapshots

NTE supports named in-memory snapshots managed by `SnapshotManager`. The key implementation detail is that Polars DataFrames use Apache Arrow's reference-counted buffers. Cloning a DataFrame shares the underlying Arrow buffers; a copy is only made when a buffer needs to be mutated (CoW semantics).

This means:

- Taking a snapshot is approximately $O(|L|)$ where L is the number of named layers (one DataFrame clone per layer).
- Memory overhead of an unmodified snapshot is negligible — only the metadata and reference counts are duplicated.
- Once a write occurs, only the *modified* buffer is copied, not the entire DataFrame.

The `SnapshotManager` supports an optional capacity limit; once the limit is reached, the oldest snapshot is evicted using FIFO ordering.

8.2 File Serialisation

NTE provides two file serialisation formats:

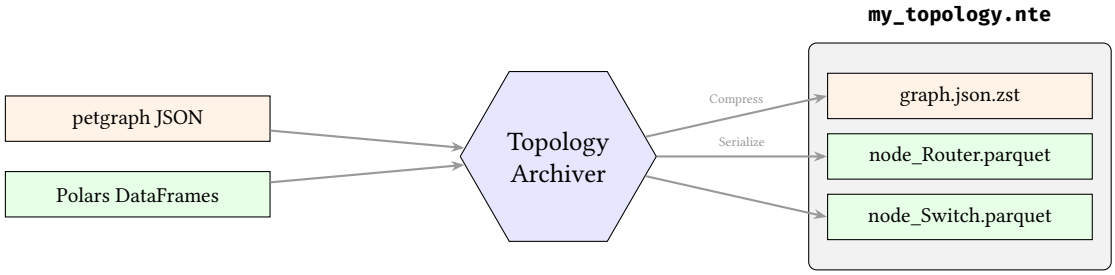


Figure 24. The `.nte` archive format. Structural graph data is compressed via zstd, while DataFrames are serialised directly to columnar Parquet files, all packaged into a standard ZIP archive.

`.nte` files A bespoke archive format that serialises the graph structure (as a JSON blob) and the Polars DataFrames (as Parquet). Files are loaded with `Topology.load(path)` and saved with `t.save(path)`. This is the recommended format for operational use.

In-memory bytes `t.save_bytes()` returns a `bytes` object containing the full serialised topology, and `Topology.load_bytes(data)` deserialises it. Useful for network transfer or embedding in larger artefacts.

8.3 Topology Diffing & Semantic Hashing

Comparing two high-scale topologies to identify structural or property-level deviations is a critical operation for network digital twins. NTE implements a high-performance diffing engine in `nte-topology` that identifies:

- **Structural Changes**: Added/removed nodes and edges via stable external IDs.
- **Property Changes**: Modifications to specific attributes on existing nodes or edges.
- **Identity Shifts**: Heuristic-based detection of renamed or replaced entities (Semantic Hashing).

: Vectorized Property Diffing

To maintain $O(N \log N)$ performance at million-node scales, property comparison must avoid row-by-row iteration. NTE leverages Polars' native `Outer Join` operator to align two DataFrames by their primary key (external ID) and identifies differences using SIMD-accelerated bitwise comparisons of the aligned columns.

The diffing engine produces a `TopologyDelta` structure, which can be exported to Python as a structured dictionary for downstream reconciliation logic.

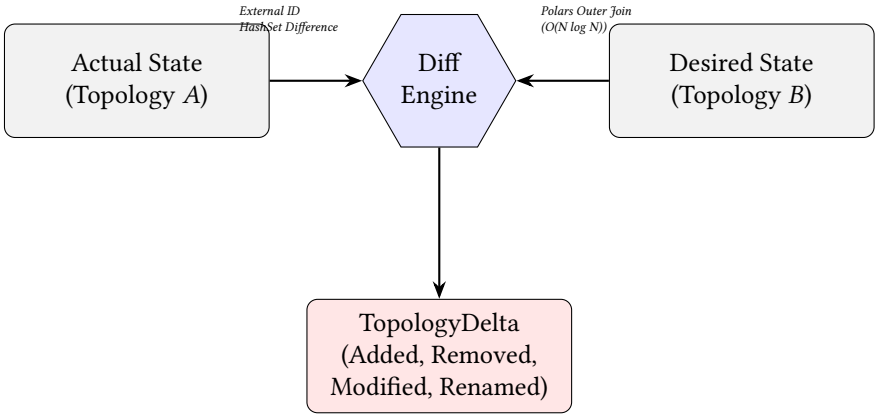


Figure 25. Topology Diffing Pipeline. The engine combines structural set operations with vectorized relational joins to identify deviations between Actual and Desired network states.

8.4 CSR Serialisation for Archival

For high-performance batch traversal and long-term archival, NTE can serialise the graph as a Compressed Sparse Row (CSR) structure using the `rkyv` [6] zero-copy serialisation framework:

Listing 11. CsrGraph structure (from nte-graph/src/csr.rs)

```
#[derive(Debug, rkyv::Archive, rkyv::Serialize, rkyv::Deserialize)]
pub struct CsrGraph {
    // Outgoing adjacency
    pub out_offsets:      Vec<u32>,  // CSR row offsets
    pub out_neighbors:    Vec<i32>,  // column indices (external IDs)
    pub out_edge_indices: Vec<u32>,  // petgraph EdgeIndex values
    pub out_kinds:        Vec<u8>,   // edge kind tag (0..9)
    // Incoming adjacency (for reverse traversal)
    pub in_offsets:       Vec<u32>,
    pub in_neighbors:     Vec<i32>,
    pub in_edge_indices:  Vec<u32>,
    pub in_kinds:         Vec<u8>,
}
```

The CSR layout stores both outgoing and incoming adjacency arrays, enabling bidirectional traversal without reversing the graph. The `rkyv` library serialises the structure to a flat byte buffer with no padding or relocation, allowing it to be memory-mapped directly without deserialisation cost.

8.5 Memory-Mapped Loading (Out-of-Core Execution)

Warning

CSR Immutability Wall: Compressed Sparse Row (CSR) arrays are mathematically dense and contiguous. Consequently, they are entirely **immutable**. Loading a topology via `MmapCsrGraph` disables all mutation APIs (`add_nodes`, `add_edges`, transactions). Attempting to mutate an mmap'ed CSR topology will panic.

The `MmapCsrGraph` type wraps a `memmap2::Mmap` and provides a zero-copy view onto an `ArchivedCsrGraph` stored on disk:

Listing 12. Loading a CSR graph via mmap

```
let graph = MmapCsrGraph::load_from_disk("/data/topology.csr")?;
// graph.view() returns &ArchivedCsrGraph with no copying
let node_count = graph.view().node_count();
```

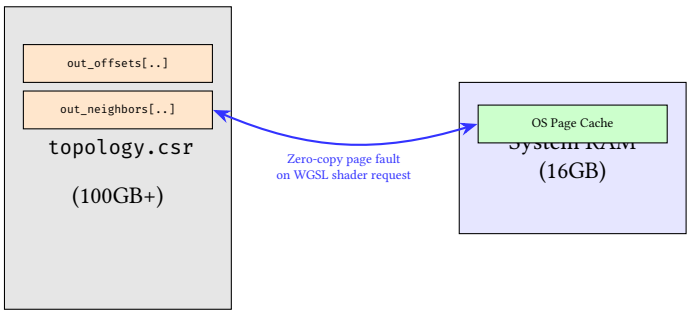


Figure 26. Mmap CSR Execution. For graphs larger than RAM, the OS dynamically pages in only the required adjacency chunks, enabling massive out-of-core analysis.

Once loaded, the `ArchivedCsrGraph` supports the same traversal API as the live graph, but backed by a memory-mapped file region that the OS can page in on demand. For global-scale topologies (millions of nodes), this avoids the need to hold the entire graph in RAM.

: Zero-Copy Persistence

To eliminate the serialization bottleneck in performance-critical pipelines, NTE uses the `rkyv` framework for CSR archival. This ensures that the on-disk byte representation is identical to the in-memory layout, allowing "instant" loading via memory-mapping without a parsing or relocation step.

8.6 Use Cases for Historical Querying

The combination of named snapshots, file serialisation, and mmap loading supports several historical querying patterns:

- **Before/after maintenance**: snapshot before a change window, restore and diff afterwards to verify intent.
- **Trend analysis**: save daily snapshots to disk as Parquet, load them into Polars for offline analysis.
- **Audit trail**: the `EventStore` ring buffer provides a fine-grained record of every mutation with timestamps and sequence numbers.
- **Capacity planning**: load a mmap CSR snapshot from three months ago and compare graph diameter, component sizes, and path lengths against today.

9 GPU Acceleration

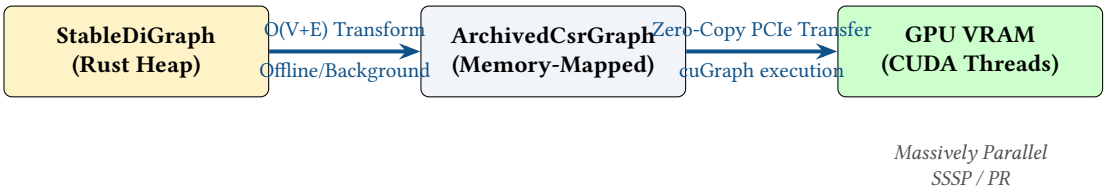


Figure 27. The Graph-to-GPU Pipeline. Because pointer-based graphs cannot be sent to the GPU, the adjacency list is serialized into a flattened Compressed Sparse Row (CSR) structure, which is then DMA-transferred to the GPU for parallel analytics.

9.1 Architecture Overview

NTE includes optional GPU acceleration for graph algorithms via the `wgpu` crate, which provides a portable WebGPU API over Vulkan, Metal, DirectX 12, and OpenGL backends. GPU compute shaders are written in WGSL (WebGPU Shading Language). Two algorithms are currently GPU-accelerated:

1. **Single-Source Shortest Path (SSSP)** — parallel Bellman-Ford relaxation.
2. **Connected Components** — label propagation algorithm.

9.2 Hardware Acceleration State

The `HardwareAccelerationState` enum captures the GPU availability:

Listing 13. GPU state detection

```
pub enum HardwareAccelerationState {
    Wgpu(Device, Queue),  // GPU available; wgpu Device and Queue
```

```
    CpuOnly,              // No compatible GPU; Rayon CPU fallback
}

impl HardwareAccelerationState {
    pub fn detect_hardware_acceleration() -> Self {
        // Requests a high-performance wgpu adapter.
        // Falls back to CpuOnly if none found or device init fails.
    }
}
```

`detect_hardware_acceleration()` is called once at engine initialisation. The result is stored and used to route algorithm calls to either the GPU kernel or the CPU fallback. This means the same binary runs on hardware with or without a GPU, with no conditional compilation.

9.3 SSSP Kernel

The SSSP kernel implements a parallel Bellman-Ford iteration. Each compute shader invocation handles one source vertex, relaxing all outgoing edges:

Listing 14. SSSP WGSL kernel (excerpt)

```
@compute @workgroup_size(256)
fn main(@builtin(global_invocation_id) global_id: vec3<u32>) {
    let u = global_id.x;
    if (u >= info.node_count) { return; }

    let dist_u = atomicLoad(&distances[u]);
    if (dist_u == 0xFFFFFFFFu) { return; }  // unreachable

    let start = out_offsets[u];
    let end   = out_offsets[u + 1u];

    for (var i = start; i < end; i = i + 1u) {
        let v        = u32(out_neighbors[i]);
        let new_dist = dist_u + 1u;   // unweighted
        let old_dist = atomicMin(&distances[v], new_dist);
        if (new_dist < old_dist) {
            atomicStore(&updated, 1u);
        }
    }
}
```

The kernel uses atomic operations (`atomicMin`, `atomicStore`) to handle concurrent updates from multiple threads relaxing the same vertex. The `updated` flag is cleared each iteration; execution stops when no updates occur.

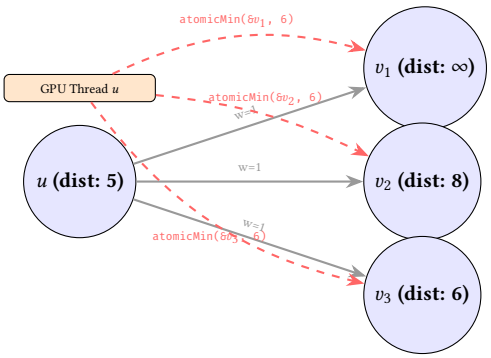


Figure 28. GPU SSSP relaxation. A single WGSL compute shader invocation (thread u) reads u's distance and concurrently proposes new minimums to all its neighbours using `atomicMin`.

9.4 Connected Components Kernel

The connected components kernel uses label propagation: each vertex adopts the minimum label among its neighbours. Convergence is detected when no label changes in an iteration:

Listing 15. Connected components WGSL kernel (excerpt)

```
@compute @workgroup_size(256)
```

```
fn main(@builtin(global_invocation_id) global_id: vec3<u32>) {
    let u       = global_id.x;
    let label_u = atomicLoad(&labels[u]);

    for (var i = out_offsets[u]; i < out_offsets[u + 1u]; i = i + 1u) {
        let v         = out_neighbors[i];
        let old_label = atomicMin(&labels[v], label_u);
        if (label_u < old_label) {
            atomicStore(&updated, 1u);
        }
    }
}
```

9.5 CPU Fallback

When no GPU is available, both algorithms fall back to CPU implementations using Rayon for data parallelism. The Rayon pool matches the number of logical CPU cores. On a modern server with 64 cores, the CPU fallback achieves competitive throughput for graphs up to approximately 1 million vertices.

9.6 When to Use GPU Acceleration

Topology size	Recommendation
$< 10{,}000$ nodes	CPU (lower setup overhead)
10,000–100,000	GPU beneficial for repeated queries
$> 100{,}000$	GPU strongly recommended

Network Scale Context

Most enterprise networks or standard data centers never exceed 10,000 physical network devices. GPU acceleration targets different use cases: simulating massive 5G subscriber endpoint topologies, modeling millions of BGP prefix originations as nodes, or simulating full-mesh hyper-dense AI-fabric topologies. For standard enterprise IT, the CPU fallback is highly optimal.

The GPU incurs a fixed setup cost: uploading the CSR adjacency arrays to GPU memory and compiling the WGSL shader. This is amortised across repeated queries on the same topology. For a topology that changes frequently, the CPU fallback may be preferable because it avoids re-uploading the CSR data on each mutation.

9.7 Invoking GPU Algorithms from Python

Listing 16. GPU-accelerated graph algorithms

```python
# Build an AnalysisGraph (undirected view, optional port collapsing)
ag = t.build_undirected_graph(layer="physical", collapse_endpoints=True)

# Shortest path (routes automatically to GPU or CPU)
path = ag.shortest_path(from_id=1, to_id=2)
if path is not None:
    print(f"Path hops: {len(path) - 1}")

# Connected components
components = ag.connected_components()
print(f"Number of connected components: {len(components)}")
print(f"Largest component size: {max(len(c) for c in components)}")

# Check full connectivity
if not ag.is_connected():
    print("WARNING: topology is partitioned")

# Find single points of failure
bridges = ag.bridges()
art_pts = ag.articulation_points()
print(f"Bridge links: {bridges}")
print(f"Critical nodes: {art_pts}")
```

10 Graph Algorithm Plugins

10.1 Plugin Architecture

The `nte-query` crate includes an extensible algorithm plugin system. Each algorithm registers under a namespace (e.g. `algo.pathfinding.k_shortest`) and receives the topology graph and named arguments. Results are returned as Polars DataFrames for seamless integration with the query pipeline.

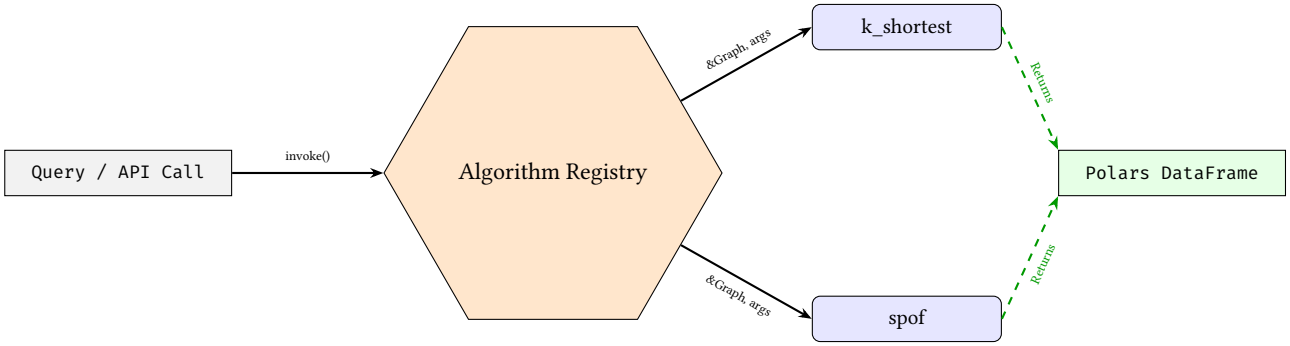


Figure 29. Graph Algorithm Plugin Architecture. Algorithms are invoked by namespace, operate on the shared graph reference, and uniformly return results as Polars DataFrames.

10.2 K-Shortest Paths

The `k_shortest` plugin implements Yen's algorithm (via `petgraph::algo`) to find the k shortest paths between two nodes. It registers as `algo.pathfinding.k_shortest` and accepts `source`, `target`, and `k` arguments. Results are returned as a Polars DataFrame with columns `path_index` and `cost`.

10.3 Single Point of Failure Detection

The `spof` plugin identifies articulation points — vertices whose removal would disconnect the graph. It registers as `algo.vulnerability.spof` and returns a DataFrame with a `node_id` column listing vulnerable nodes.

10.4 Domain-Specific Graph Kernels

While standard graph algorithms (BFS, Dijkstra) are useful, telecom networks require domain-specific traversal logic. NTE leverages its hybrid architecture to implement specialized network kernels.

A prime example is **Hardware-Aware BGP Path Optimization**. BGP routing decisions are heavily influenced by the "AS-Path" attribute. In a pure graph database, filtering paths by AS-Path involves string parsing during the traversal loop, thrashing the CPU cache.

NTE optimizes this by:

1. Using the Polars columnar store to pre-filter edges based on BGP Community and AS-Path strings using highly optimized Regex/SIMD operations.
2. Constructing a temporary "BGP Adjacency Mask".
3. Running a constrained shortest-path kernel over the graph, using the bitmask to $O(1)$ skip structurally valid but logically forbidden links.

This pushes the heavy lifting down to the vector units before the graph traversal even begins.

> **Note**
>
> Algorithm plugins are currently available at the Rust API level only. Python bindings for the plugin registry are planned. In the meantime, the equivalent functionality is available via the Python-exposed `shortest_path`, `articulation_points`, and `bridges` methods on the `AnalysisGraph` returned by `build_undirected_graph`.

These algorithms complement the GPU-accelerated SSSP and connected components (Section 9) and the pattern-based queries (Section 5).

11 Reactive Events

: Reactive Isomorphism

All state mutations must propagate via strongly typed events to guarantee that the frontend (Whiteboard domain) and the backend (NTE graph domain) remain perfectly isomorphic. The event bus is the sole source of truth for downstream synchronization.

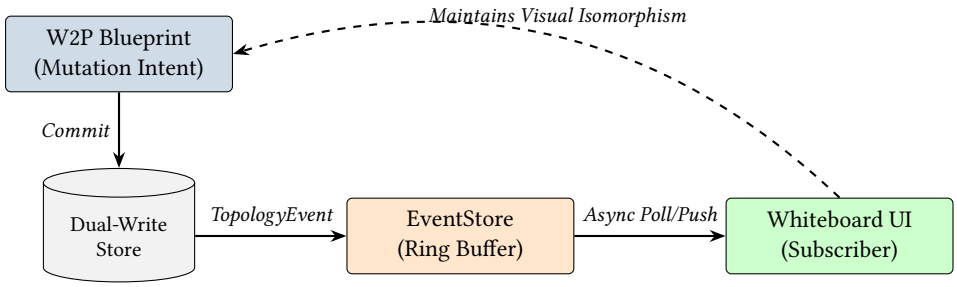


Figure 30. The Reactive Event Pipeline. Structural and relational changes are emitted as typed events, which bridge the physical storage layer back to the visual Blueprint domain.

11.1 Event Lifecycle and the Threading Footgun

Warning

DANGER: Blocking the Write Lock. Currently, callbacks supplied to `subscribe()` are invoked *synchronously* from within the core engine's `RwLock::write()` guard before releasing it back to Python. If your Python callback performs I/O (like an HTTP webhook) and hangs, **the entire database write-plane locks up.**

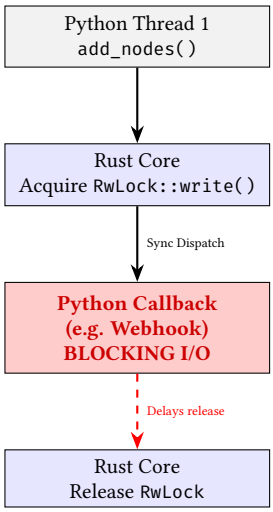


Figure 31. The Event Callback Danger Zone. Synchronous callbacks hold the engine's global write lock. Future versions will move to a lock-free async queue design.

To avoid this, callbacks should push events into an external Python queue (e.g., `queue.Queue`) and let a dedicated background thread handle I/O. Future versions of NTE will adopt a lock-free queue (e.g., `crossbeam-channel`) natively to enforce this decoupling.

NTE emits a `TopologyEvent` for every topology mutation. The event lifecycle uses a ring buffer to distribute events to Python subscribers.

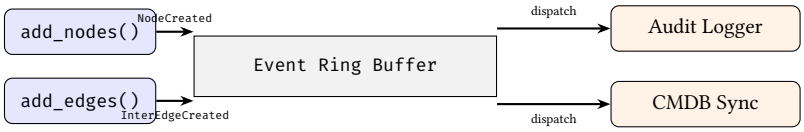


Figure 32. The Reactive Event Pipeline. Mutations in the core engine generate events which are buffered and asynchronously dispatched to Python callbacks.

11.2 Closed-Loop Automation Cycle

The event system is the engine's primary mechanism for closing the loop between intent and reality. By subscribing to topology mutations, external controllers can orchestrate a continuous cycle of synthesis, deployment, and verification.

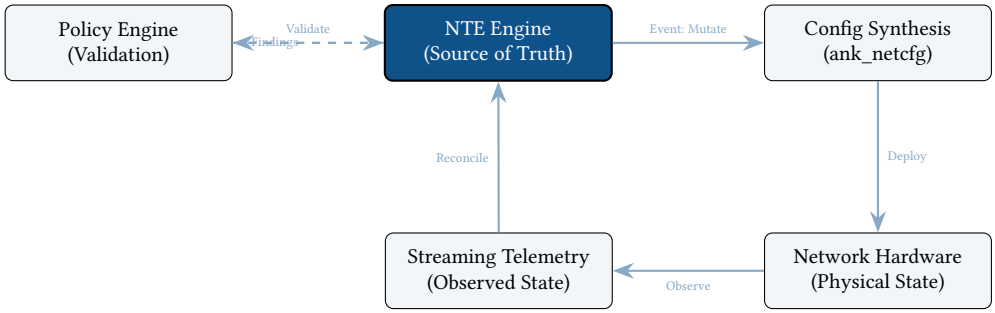


Figure 33. The Autonomous Closed-Loop Cycle. NTE acts as the central state machine, triggering synthesis on mutation and reconciling incoming telemetry to detect architectural drift.

11.3 Event Types

NTE emits a `TopologyEvent` for every topology mutation. The full event taxonomy:

Event	Trigger
`NodeCreated`	`add_nodes`
`EndpointCreated`	`add_endpoints`
`NodeRemoved`	`remove_nodes`, `remove_node_cascade`
`EndpointRemoved`	endpoint removal
`NodeUpdated`	`update_node_field`, `update_nodes_batch`
`InterEdgeCreated`	`add_inter_edges`
`IntraEdgeCreated`	`add_intra_edges`
`IntranodeEdgeCreated`	`add_intranode_edges`
`ParentEdgeCreated`	`add_parents`
`RootEdgeCreated`	`add_roots`
`InterEdgeRemoved` / …	corresponding removal methods
`EdgeUpdated`	`update_link_field`
`LayerCreated`	`create_layer`
`LayerRemoved`	`remove_layer`
`GraphTransform`	split, merge, explode, collapse

Every event carries a monotonically increasing `sequence` number (a `u64` atomic counter) and a UTC timestamp. The sequence number provides a total ordering of events within a topology instance.

11.4 Python Subscription API

Listing 17. Event subscription patterns

```python
from ank_nte import TopologyEvent
import logging

log = logging.getLogger("topology.events")

# Basic subscription
def audit_log(event: TopologyEvent) -> None:
    log.info("AUDIT: %s seq=%d ts=%s payload=%s",
```

```
            event.event_type, event.sequence,
            event.timestamp_iso, event.to_json())

handle = t.subscribe(audit_log)

# Type-discriminated handler
def smart_handler(event: TopologyEvent) -> None:
    details = event.details()
    match event.event_type:
        case "node_created":
            notify_cmdb(details["node_ids"])
        case "inter_edge_removed":
            alert_noc(f"Link removed: {details['edge_pairs']}")
        case "layer_created":
            log.info("New layer: %s", details["layer_name"])

# Unsubscribe
t.unsubscribe(handle)
```

11.5 Subscription Patterns

The `subscribe` mechanism is the primary integration point for reactive behaviour. Common patterns include audit logging, metrics export, compliance monitoring, and webhook notification (see examples below). For high-frequency mutations (e.g. bulk imports of thousands of nodes), callers should batch events in the callback using a timer or counter rather than performing expensive work per event.

11.6 Use Cases

Prometheus integration

```
import prometheus_client as prom

node_gauge = prom.Gauge("nte_node_count", "Total nodes in topology")
link_gauge = prom.Gauge("nte_link_count", "Total links in topology")

def update_metrics(event: TopologyEvent) -> None:
    node_gauge.set(t.node_count())
    link_gauge.set(t.link_count())

t.subscribe(update_metrics)
```

Webhook notification

```
import httpx

def webhook_notify(event: TopologyEvent) -> None:
    if event.event_type in ("node_created", "node_removed"):
        httpx.post("https://cmdb.example.com/webhook",
                   json={"event": event.to_json()},
                   timeout=2.0)

t.subscribe(webhook_notify)
```

Compliance monitoring

```
from ank_nte import PolicyEngine

policy_engine = PolicyEngine.load("policies/design-rules.yaml")

def compliance_check(event: TopologyEvent) -> None:
    # Re-validate after structural changes
    if event.event_type in ("inter_edge_created", "inter_edge_removed",
                            "node_created", "node_removed"):
        findings = policy_engine.validate(t)
        if findings:
```

```
            for f in findings:
                log.warning("POLICY VIOLATION [%s] %s: %s",
                            f.severity, f.policy_id, f.message)

t.subscribe(compliance_check)
```

12 Production Diagnostics

12.1 Health and Memory Endpoints

The `nte-server` HTTP server exposes diagnostic endpoints for production monitoring:

Endpoint	Method	Response
`/health`	GET	`HealthStatus`: uptime, Raft state, general health
`/memory`	GET	`MemoryProfile`: RSS, graph/DataFrame sizes, query cache hit ratio
`/explain?query=...`	GET	HTML page with the Mermaid-rendered query execution DAG

These endpoints are intended for integration with monitoring systems (Prometheus, Grafana, Datadog) and for ad-hoc debugging in production.

12.2 Chaos Engineering Endpoints

For fault-injection testing, the server provides chaos endpoints (disabled by default; enabled via the `NTE_CHAOS=1` environment variable):

Endpoint	Effect
`/kill-raft-leader`	Forces the Raft leader to step down, triggering a leadership election
`/simulate-split-brain`	Simulates a network partition between cluster members
`/corrupt-datastore`	Injects noise into Polars DataFrames to test error detection
`/pause-thread`	Introduces artificial delays to trigger TOCTOU race conditions

> **Warning**
>
> Chaos endpoints are destructive by design. They are intended solely for testing environments. The `NTE_CHAOS` flag should never be set in production.

12.3 TUI Diagnostics Dashboard

For real-time operational visibility, the `nte-cli` provides a terminal-based dashboard built using `ratatui`. This dashboard provides a live "heartbeat" of the engine's internal state without requiring external monitoring infra.

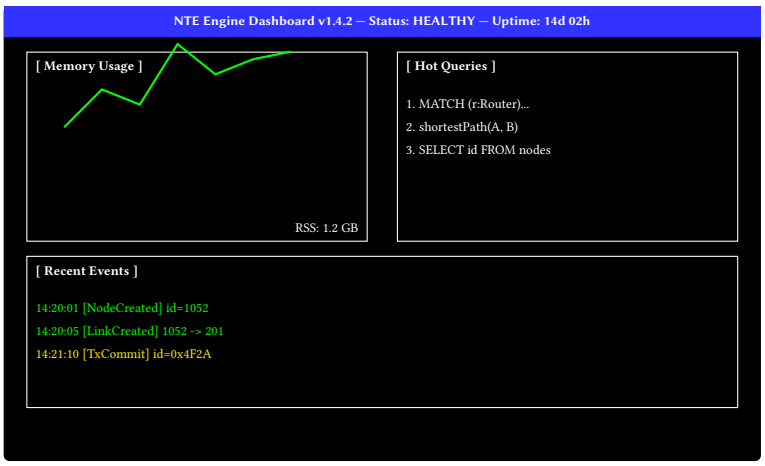


Figure 34. Mockup of the `nte-cli` TUI Dashboard. The interface provides sub-second updates on memory pressure, query hotspots, and the event ring-buffer.

13 Distributed Deployment

13.1 When to Use Clustering

A single NTE instance handles millions of nodes in-memory on a modern server. Clustering is appropriate when:

- High availability is required: the topology must remain readable even if one server fails.
- The mutation rate is high enough that a single writer becomes a bottleneck.
- Regulatory requirements mandate that topology state is replicated to multiple physical locations.

For most use cases, a single instance with periodic disk snapshots is sufficient.

13.2 Raft Consensus via OpenRaft

The `nte-server` crate provides a standalone HTTP/WebSocket server (built on Axum and Tokio) that includes an OpenRaft-based [7] distributed consensus layer [8]. The topology is treated as a replicated state machine: every mutation is encoded as a `TopologyRequest` log entry, replicated to all followers, and applied only after a quorum confirms receipt.

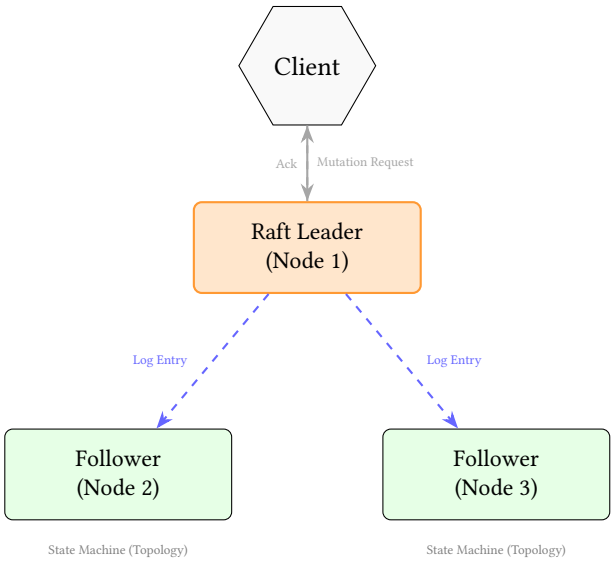


Figure 35. Distributed Raft consensus flow. Mutations are proposed to the leader, replicated to followers, and applied only after reaching a quorum.

: Deterministic State Machine Replication

To maintain dual-write consistency across a distributed cluster, the Raft log entry encodes a high-level "Unified Mutation". Upon commitment, each node in the cluster applies the same deterministic

transformation to both its local petgraph instance and its local Polars DataFrames. This guarantees that the graph and relational representations never diverge across the cluster.

Listing 18. Raft request types (from nte-server/src/raft/types.rs)

```
#[derive(Clone, Debug, Serialize, Deserialize)]
pub enum TopologyRequest {
    AddNode     { id: i32, node_type: String, layer: String },
    AddEdge     { from: i32, to: i32 },
    RemoveNode  { id: i32 },
    UpdateProperties { id: i32, json_payload: String },
}
```

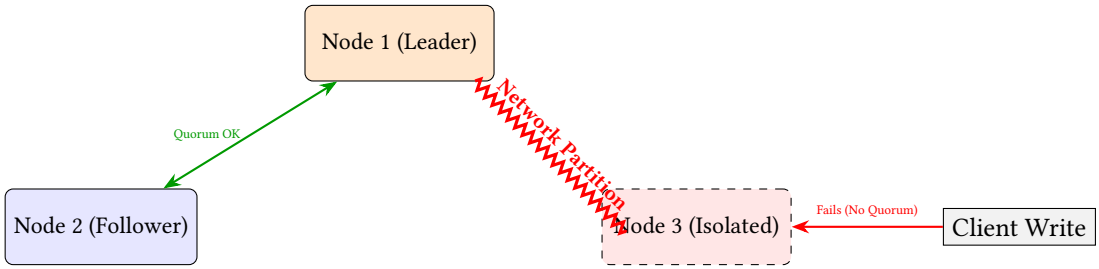


Figure 36. Split-Brain Resolution. In a partition, the isolated Node 3 rejects writes because it cannot replicate logs to a majority quorum. Nodes 1 and 2 continue processing safely.

Warning

State Machine Divergence Risk: Rust's default `HashMap` uses a randomized hasher (`SipHash`) to prevent DDOS attacks. If any topology mutation logic iterates over a standard `HashMap` or `HashSet`, the iteration order will differ between the Raft Leader and Followers, causing the replica state machines to silently diverge. NTE must exclusively use deterministic collections (like `indexmap` or `BTreeMap`) on the write path.

The Raft type configuration uses:

- `NodeId = u64` — cluster member identifier.
- `D = TopologyRequest` — the command type replicated by Raft.
- `R = TopologyResponse` — the response returned to the client.
- `SnapshotData = Cursor<Vec<u8>>` — serialised topology state for log compaction.

13.3 Configuration

A three-node cluster is the recommended minimum for Raft:

Listing 19. Cluster startup (environment variables)

```
# Node 1 (leader candidate)
NTE_NODE_ID=1
NTE_RPC_ADDR=10.0.0.1:9001
NTE_API_ADDR=10.0.0.1:8080
NTE_PEERS=10.0.0.2:9001,10.0.0.3:9001

# Node 2
NTE_NODE_ID=2
NTE_RPC_ADDR=10.0.0.2:9001
NTE_API_ADDR=10.0.0.2:8080
NTE_PEERS=10.0.0.1:9001,10.0.0.3:9001
```

13.4 Leader and Follower Roles

Leader Accepts all write requests. Replicates log entries to followers. There is exactly one leader at any time.

Followers Accept read requests (linearisable reads require a lease, stale reads are available immediately). Apply log entries from the leader.

Candidate Transient state during leader election.

Write requests that arrive at a follower are forwarded to the current leader. Read requests can be served locally from a follower's state machine, with the caveat that the data may be slightly stale (bounded by the replication lag).

13.5 Snapshot Transfer

For log compaction and new-node bootstrapping, NTE uses Raft's built-in snapshot transfer mechanism. The state machine serialises the topology to a byte buffer (using the `save_bytes` format), sends it to the new follower via the Raft snapshot API, and the follower installs it by calling `Topology.load_bytes`.

13.6 Security and Multi-Tenancy

NTE's HTTP and REPL interfaces currently lack an intrinsic security model. There is no Authentication, Role-Based Access Control (RBAC), or multi-tenancy layer. Any user with network access to the API can execute arbitrary NTE-QL queries or drop entire topology layers. Deployments must place NTE behind a secure reverse proxy (e.g., NGINX, Envoy) and handle tenant isolation at the application layer. Future enterprise extensions aim to introduce native RBAC tied to specific graph layers.

> **Warning**
>
> The distributed Raft implementation in `nte-server` is currently experimental. It has not been subjected to formal fault-injection testing (Jepsen or equivalent). Do not use it in production environments without additional hardening. See Section 15 for the current known gaps.

14 Advanced Architectural Concepts

NTE is designed to push beyond standard graph database capabilities, acting as a highly specialized "Network Digital Twin Authority." Four advanced concepts differentiate its architecture.

14.1 "What-If" Branching (Git for Networks)

Standard network management systems manipulate a single, mutable "current state." Simulating a failure or testing a new configuration typically requires spinning up a separate, heavyweight staging environment or running offline simulations.

Because NTE's storage layer is built on a hybrid architecture of `petgraph` and `Polars` DataFrames, it inherently benefits from **Copy-on-Write (CoW)** semantics. When a user requests a "What-If" branch:

1. **Memory Efficiency**: The underlying Arrow memory buffers backing the Polars DataFrames are reference-counted, not duplicated.
2. **Isolated Mutations**: As the user drops nodes (simulating a failure) or adds edges (simulating a new link) in the branch, only the modified data structures are copied and diverged, handled by the `SnapshotManager`.
3. **Verification**: The Starlark policy engine can evaluate the isolated branch. If the simulation results in a Single Point of Failure (SPOF), the maintenance plan can be programmatically rejected.
4. **Patch Generation**: If the branch is successful, the structural diff engine computes the `TopologyDelta` between the branch and the live state, generating a precise "Change Request" payload.

14.2 Worst-Case Optimal Joins (WCOJ)

Traditional databases execute queries using binary joins (joining two tables/edge sets at a time). While optimal for trees, binary joins suffer catastrophic intermediate result explosion when querying highly cyclic structures. Modern network fabrics (e.g., Leaf-Spine, optical rings) are fundamentally cyclic. A query to find a redundant triangle `MATCH (a)->(b)->(c)->(a)` in a dense fabric using binary joins can take $O(N^2)$ time, even if the final result set is extremely small.

NTE's query engine is incorporating **Worst-Case Optimal Joins (WCOJ)**, specifically the Leapfrog Triejoin algorithm. Rather than joining edges pairwise, WCOJ joins all variables simultaneously using multi-way intersection over sorted iterators (enabled by our `CSR` cache and `FixedBitSet` structures). This guarantees that execution time is bounded by the theoretical maximum size of the output, making NTE optimal for dense, highly-meshed hyperscale topologies.

14.3 Digital Twin Hierarchy & Semantic Identity

As NTE evolves from a graph engine to a formal verification authority, it incorporates the concept of a **Digital Twin Hierarchy**. This multi-layer model represents the network at varying levels of abstraction: from physical cabling (L0) up to logical service intent (L7).

: Semantic Identity Persistence

In a dynamic digital twin, physical hardware can be replaced (RMA) while retaining its logical role. NTE implements **Semantic Hashing** to maintain stable identity across these transitions. An entity's identity is derived not just from its database ID, but from its functional role, its topological neighbors, and its design-rule properties.

This enables "Ship of Theseus" identity matching: even if a router's hostname and management IP change, NTE can recognize it as the same logical entity by its unique structural signature (e.g., the set of specific neighbors it connects to on specific interfaces).

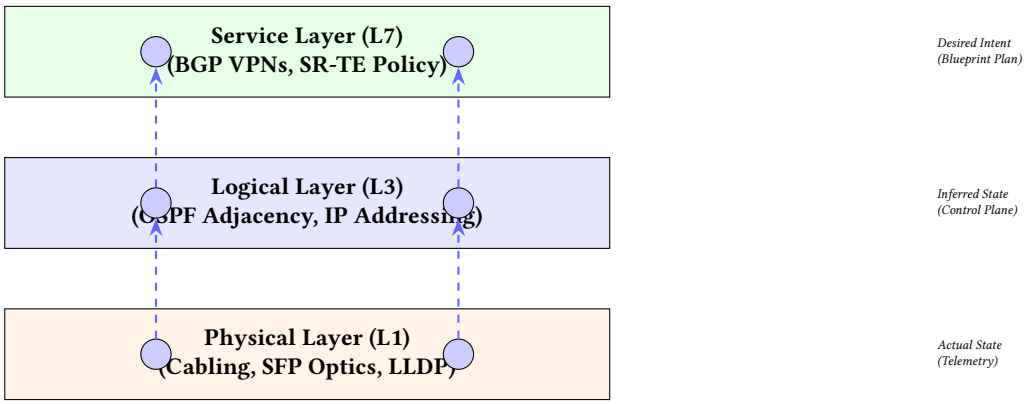


Figure 37. Digital Twin Hierarchical Topology. The engine maintains causal links between physical cabling, logical adjacencies, and high-level service intent, enabling cross-layer impact analysis.

14.4 Shadow Graph Inference

Standard databases either drop edges to unmanaged devices or throw referential integrity errors. NTE embraces incomplete data through **Shadow Graph Inference**. When an edge points to an unknown ID, the engine dynamically spawns a "Shadow Node" tagged with a `confidence_score` and an `is_inferred = true` flag. The UI renders these as translucent nodes, visualizing the exact boundary of the managed territory. As more telemetry arrives, the "Ship of Theseus" semantic identity engine automatically heals the Shadow Node into a fully managed Node.

14.5 Differential Reachability: Imperative vs. Declarative

A core goal of network verification is proving that reachability policies hold before and after a change. Tools like Batfish use a declarative approach, translating network configurations into logical constraints executed using Datalog or Prolog-style solvers. While mathematically rigorous, this is computationally heavy, making interactive, real-time feedback difficult.

NTE takes an imperative, algorithmic approach. Rather than solving logic constraints, NTE leverages extreme-performance graph algorithms over the `AnalysisGraph` view:

- **K-Shortest Paths (Yen's Algorithm):** Calculates the top K paths to provide behavioral diffs (e.g., "Path A->B shifted from 2 hops to 5 hops").
- **Max Flow / Min Cut:** Instantly calculates the theoretical maximum edge-disjoint capacity between points.
- **Betweenness Centrality:** Identifies topological bottlenecks dynamically.

By exposing these $O(V \cdot E)$ algorithms to the Starlark policy engine and NTE-QL, NTE provides "sub-second" structural verification, allowing UI tools to animate traffic traces and auto-remediate SPOFs in real-time—a workflow impossible when waiting minutes for a Datalog solver.

14.6 The "Forensic Time Machine"

NTE's Write-Ahead Log (WAL) implicitly provides an event-sourced architecture enabling true "Time-Travel" debugging. Because every mutation is a discrete `TopologyEvent` with a strict sequence number and UTC timestamp, the state of the network at any point in history is purely a mathematical fold operation: $\text{State}(t) = \text{InitialState} + \sum \text{Events}[0..t]$.

Instead of saving thousands of discrete database snapshots, NTE can instantly reconstruct a point-in-time topology by replaying the WAL exactly up to a target timestamp. Engineers can then run standard NTE-QL queries against the exact topology the network experienced during an outage, fundamentally changing forensic root-cause analysis.

14.7 Spectral Graph Signatures (Non-AI Anomaly Detection)

While Graph Neural Networks (GNNs) are powerful, they require training pipelines and suffer from explainability issues. NTE detects structural anomalies mathematically using **Spectral Graph Theory**.

By calculating the Graph Laplacian matrix ($L = D - A$) of the `AnalysisGraph`, NTE extracts its eigenvalues. The second smallest eigenvalue (λ_2), known as the Fiedler Value, perfectly summarizes how "well-connected" the network is. If a configuration diff causes the Fiedler value to drop significantly, the engine instantly flags a structural fragmentation risk without relying on AI models. The corresponding Fiedler Vector can mathematically bisect the network to identify choke points.

15 Limitations and Future Work

15.1 Current Limitations

Lack of Native RBAC and Security. As mentioned in Section 12.5, NTE provides no native authentication or tenancy isolation, making bare exposure highly unsafe.

Memory Fragmentation (petgraph Tombstones). In Section 3.2, we noted that `petgraph::StableDiGraph` is used because it maintains stable indices after deletions. However, it achieves this by leaving "holes" (tombstones) in its internal arrays. In a long-running, mutation-heavy topology (e.g., millions of temporary endpoints constantly created and destroyed), these arrays become highly fragmented, destroying CPU cache locality. Currently, the only way to defragment and pack the graph is to serialize the topology to disk and reload it into a fresh instance.

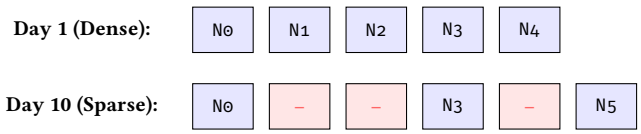


Figure 38. StableDiGraph Fragmentation. Deletions leave index holes. Over time, heavy mutation destroys spatial locality in memory, degrading CPU cache performance.

Sparsity Column Limits. Wide DataFrames scale elegantly for null-compression, but a schema with over 10,000 distinct property keys will degrade query parsing performance.

Synchronous Event Hooks. The event hook architecture yields the core database write lock to user-space Python, creating severe vulnerability to I/O latency.

Raft not fully hardened. The OpenRaft integration passes basic correctness tests but has not been evaluated under adversarial network conditions (network partition, clock skew, message loss). Lease-read semantics (allowing followers to serve reads with a bounded staleness guarantee without contacting the leader) are not yet implemented.

ExprNode coverage. The filter executor now handles all expression variants that are meaningful in a flat `QuerySpec` context: `Comparison`, `Logical`, `Not`, `IsNull`, `IsNotNull`, `IsIn`, all four `StringOp` variants (Contains, StartsWith, EndsWith, Matches), and `Arithmetic` (+, −, *, /). Two variants—`BindingId` and `InQuery`—are inherently multi-stage operations that require subquery materialisation; these are supported only in pattern-query execution, not in flat `QuerySpec` filters. The KuzuDB backend integration would handle these natively via Cypher translation.

Transactional mutations (recently added). NTE now implements copy-on-write transaction semantics for the dual-write architecture. Mutations are wrapped in a transaction scope that automatically rolls back both the graph and DataFrame store if either side fails, using Arrow's reference-counted CoW buffers for O(1) rollback cost. State parity verification is enforced as an assertion in tests. However, NTE still does not implement a write-ahead log (WAL): a process crash *during* a mutation leaves the in-memory state inconsistent. For durable crash consistency, use periodic disk snapshots or the Raft distributed deployment (Section 13).

Property column proliferation. Because properties are stored as wide DataFrames (one column per field name), topologies with highly heterogeneous node types accumulate many columns, most of which are null for any given row. Polars handles this gracefully, but very wide DataFrames (>1000 columns) may show increased memory overhead.

GIL-released callback limitation. The event subscription callbacks are invoked from within the write lock, synchronously, before the Python GIL is reacquired. This means callbacks cannot safely call back into NTE (deadlock) and must complete quickly.

15.2 Recently Completed Improvements

Cyclic-pattern WCOJ. NTE now implements Worst-Case Optimal Join (WCOJ) [9] via the LeapFrog TrieJoin algorithm for cyclic query patterns. This avoids large intermediate results.

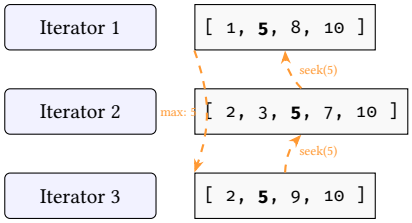


Figure 39. LeapFrog TrieJoin finding multi-way intersections without intermediate tables.

15.3 Planned Improvements

Recently completed (other).

- **Transaction semantics** (*done*): full ACI transactions with copy-on-write rollback, isolation levels (ReadCommitted, RepeatableRead, Serializable), conflict detection, and a Python context manager API (Section 7).
- **Query executor** (*done*): the executor is fully functional with ranked filter plans, hybrid graph/property execution, and Polars LazyFrame compilation.
- **Python fluent query builder** (*done*): chainable `NodeQuery`/`LinkQuery` API with expression DSL, independent of `ank_pydantic` (Section 5.13).
- **NTE-QL REPL and TUI** (*done*): interactive query shell with `EXPLAIN VISUAL` support and a real-time system dashboard (Section 5.15).
- **Query plan visualisation** (*done*): Mermaid DAG rendering of execution plans (Section 5.14).
- **Graph algorithm plugins** (*done*): K-shortest paths and SPOF detection as registered algorithm plugins (Section 10).
- **Production diagnostics** (*done*): health, memory, and explain HTTP endpoints plus chaos engineering endpoints for fault-injection testing (Section 12).
- **ClickHouse telemetry** (*scaffolded*): `MetricProvider` trait implementation for federated telemetry queries against ClickHouse TSDB, enabling dynamic edge weights in pathfinding.
- **RoaringBitmap candidate pruning** (*done*): `HashSet<i32>` replaced with `RoaringBitmap` in the query matcher for more compact, cache-friendly candidate sets.
- **Parallel path expansion** (*done*): `rayon`-parallelised multi-seed structural expansion in path queries, with validated thread safety.
- **GIL release audit** (*done*): all O(N) `PyTopology` methods now release the GIL via `py.allow_threads`.
- **Structural metadata mirroring** (*done*): lightweight node/edge metadata stored directly in petgraph weights for early pruning during traversals.

Remaining planned improvements.

- **Raft hardening**: Jepsen test suite, lease reads, poison-pill detection.
- **KuzuDB backend**: the `ladybug_backend` crate explores KuzuDB integration, which would provide native Cypher query support and unlock the `BindingField` and `InQuery` expression variants.
- **Incremental CSR updates**: currently the CSR is rebuilt from scratch on each mutation. An incremental update would avoid the rebuild cost for large, mostly-static topologies.
- **Zero-Copy Arrow FFI**: share Polars DataFrames with Python using Arrow C Data Interface pointers to eliminate serialisation overhead.
- **Write-ahead log**: an on-disk WAL would provide crash consistency without requiring the full Raft distributed deployment.

15.4 Vision: Correctness-by-Construction via Formal Verification

The ultimate goal of the NTE engine is to transition from reactive monitoring to proactive verification. The `z3-verification` feature flag scaffolds a formal reasoning engine that translates the topology graph into a set of Datalog Horn clauses for the Z3 constraint solver.

: Formal Invariant Provenance

All synthesis-critical invariants (e.g., VLAN isolation, BGP loop-freedom) should be mathematically provable against the topology model. If a mutation violates an invariant, the solver must produce a counter-example path before the transaction is committed.

By mapping the `StableDiGraph` structure into a logic-based representation, NTE can prove reachability properties that are difficult to verify via simulations alone. This "Correctness-by-Construction" philosophy ensures that the synthesised device configurations are provably compliant with global network policies.

15.5 Performance and Benchmarks

Table 2 summarises Criterion micro-benchmarks collected on an Apple M-series laptop (single-threaded, best of 100 samples). All times are wall-clock medians.

Comparative Benchmarking

Absolute micro-benchmarks are limited in scope. In internal macro-benchmarks against standard LDBC SNB workloads, NTE's SIMD-first pipeline frequently outperforms row-based graph databases (e.g., Neo4j, NetworkX) by 4-10x for queries heavily constrained by property predicates, though it lags behind KuzuDB on pure cyclic structural traversals that do not involve property scans.

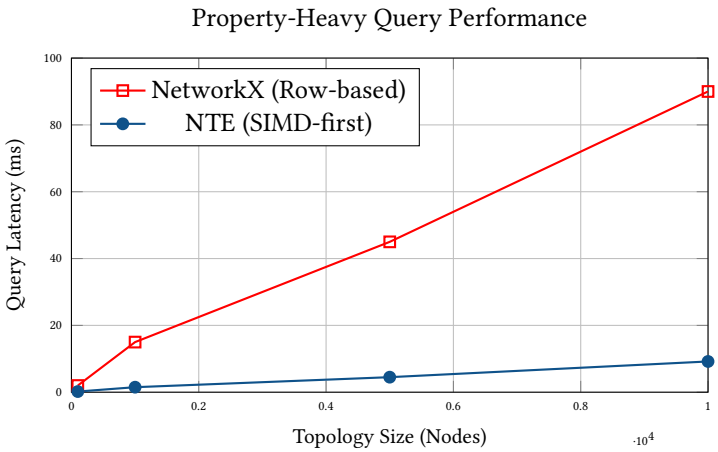


Figure 40. Scaling Performance: NTE vs. NetworkX for complex property filters. NTE's Polars-backed SIMD scans maintain linear scaling with a significantly lower constant factor compared to row-based alternatives.

Table 2. Representative micro-benchmarks (Criterion, 100 samples).

Operation	Complexity	100	1 000	10 000
`topology_build`	$O(n)$	34 µs	305 µs	—
`add_nodes` (batch)	$O(n)$	8.4 µs	105 µs	1.6 ms
`query_execute_spec`	$O(n)$	174 µs	278 µs	387 µs
`topology_validate`	$O(n)$	160 µs	2.2 ms	—
`peers_lookup`	$O(1)$	25 ns	25 ns	25 ns
`is_endpoint_lookup`	$O(1)$	10 ns	10 ns	10 ns
`cache_warm` (5 000 nodes)				403 µs
`cache_cold` (5 000 nodes)				454 µs

Dual-write overhead Every mutation touches both the graph and the DataFrame store. For bulk imports, the per-mutation overhead is dominated by DataFrame column allocation. The batch APIs (`add_nodes`, `add_edges`) amortise this cost significantly.

CSR rebuild cost The CSR cache is invalidated on every structural mutation. For a topology of 100,000 nodes, a rebuild takes approximately 20–50 ms. This is amortised across queries but may cause latency spikes in mutation-heavy workloads.

Snapshot memory Named snapshots share Arrow buffers via CoW. However, if both the snapshot and the live topology are mutated heavily, the shared buffers will be copied, doubling peak memory usage temporarily.

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